

NC STATE UNIVERSITY

ECE 533 MIDTERM PROJECT **Synchronous Buck Converter Design**

Submitted to
Prof. Dr. Douglas Hopkins
Electrical & Computer Engineering

Submitted By
Team 4

Date of Submission: 23/03/2025

Table of Contents

Academic Integrity

[Drive link](#) for project files

1. **Problem 1: Thermomechanical Material Interactions**
 - a. **Problem Statement**
 - b. **Design and Simulation Approach**
 - c. **Result Table**
 - d. **Interpretation From Results**
2. **Problem 2**
 - a. **Problem Statement**
 - b. **Design and analysis of synchronous buck converter**
 - c. **Design of physical layout of the Power Module**
 - d. **Electrical Analysis of freewheeling diode path**
 - e. **Thermal Analysis**
3. **Problem 3**
 - a. **Problem Statement**
 - b. **Design of the Planar Inductor**
 - c. **Results**

Academic Integrity

We certify that, we have neither given nor received unauthorized aid on this project.

Drive link for project files

Problem 1: Thermomechanical Material Interactions

Problem Statement:

In this problem, we have to study the effect of thermal stimulus on the mechanical properties of a cross section of a larger of a larger PCB board, comprised of multiple layers of varying material properties. These include two layers of Copper and Nano Copper on the top and bottom, which sandwich an ARC layer in the middle. A maximum t_{Cu} needs to be found for a given t_{Ncu} . This is to be done by gradually varying the porosity from 0 to 30%. The properties of Nano-Cu are scaled accordingly, for e.g.: The thermal conductivity of Nano-Cu is taken to be 95% of that of Cu for 5% porosity. Manual approximations of the equivalent thermal conductivity and electrical resistivity are included in the results based on the extrapolated results from the simulator.

Design and Simulation Approach:

(I) Setting up the simulation in Ansys:

In Ansys workbench, the steady-state thermal model is utilised to create a simple quarter-symmetry to ease the computational complexity. The dimensions are scaled down accordingly in mm for convenience. The module is further connected to the static-structural analysis environment to impart mechanical properties of the material. Width and Length of the Cu and Nano-Cu layers was set to 5mm and 15mm respectively. ARC layer was kept at 10mm X 30 mm.

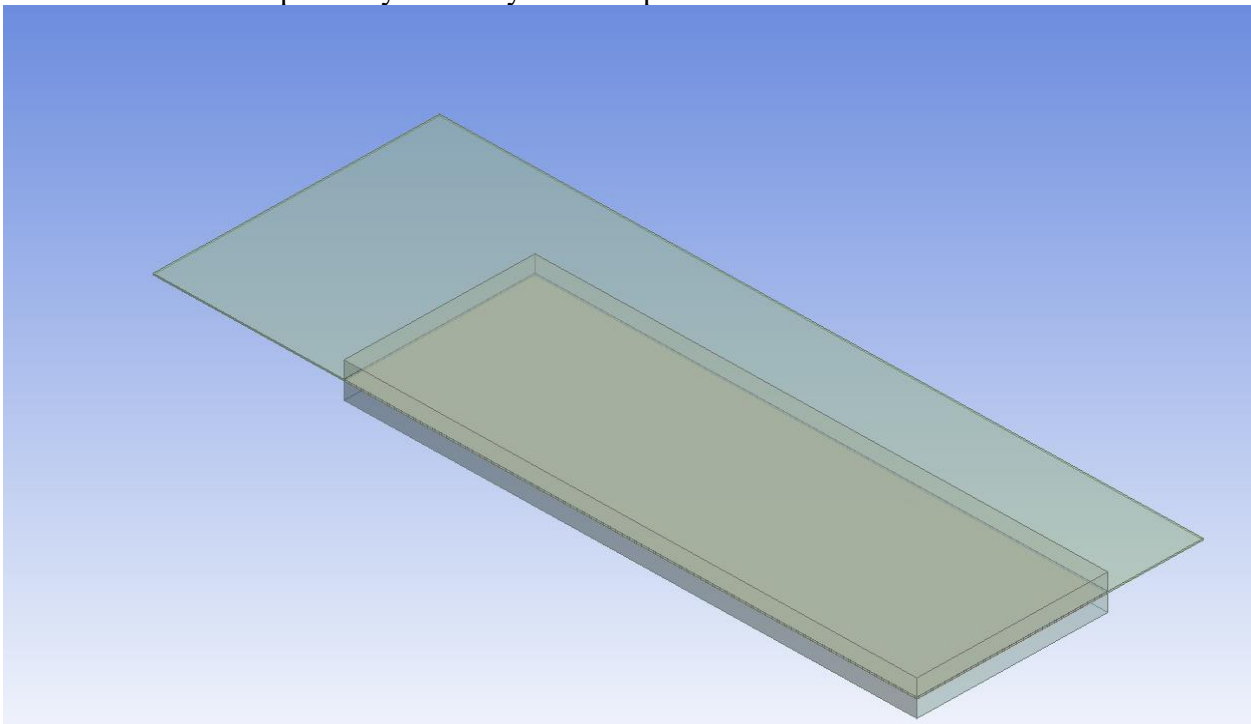


Fig 1.1: Design Modeler geometry Setup

The following engineering data was entered for the material:

Copper:

- i. CTE: 401 W/m-k
- ii. Youngs Modulus: 117×10^9 Pa
- iii. Poisson's Ratio: 0.34
- iv. Extracted Bulk Modulus: 1.218×10^{11} Pa
- v. Extracted Sheer Modulus: 4.365×10^{10} Pa

Nano-Copper:

- i. CTE: Scaled in proportion to porosity
- ii. Youngs Modulus: Scaled in proportion to porosity
- iii. Poisson's Ratio: Scaled in proportion to porosity
- iv. Extracted Bulk Modulus: Scaled in proportion to porosity
- v. Extracted Sheer Modulus: Scaled in proportion to porosity

ARC:

- i. CTE: 36 W/m-k
- ii. Youngs Modulus: 380×10^9 Pa
- iii. Poisson's Ratio: 0.23
- iv. Extracted Bulk Modulus: 2.345×10^{11} Pa
- v. Extracted Sheer Modulus: 1.544×10^{11} Pa

(II) **Setting Up the Analyses:**

Initial Assumptions:

- 1) Initial temperature: 25°C
- 2) Final temperature: 150°C
- 3) We have assumed that there are no residual stresses in the components.
- 4) Quarter of the total symmetry.

From the geometry, all the parts were combined into a single unit. Respective material assignment was provided to the materials. Meshing across all the bodies was selected to be of an automated nature with a default element size. Very fine meshing increases the computational complexity. With an initial temperature condition of 22°C, various temperatures were given to the model (50 to 150°C) to test the stress-strain relationship and evaluate the breaking point of the model. The analyses conducted include:

- i. Total Heat Flux (Thermal)
- ii. Total Deformation (Structural)
- iii. Maximum Principal Stress (Structural)
- iv. Maximum Principal Elastic Strain (Structural)

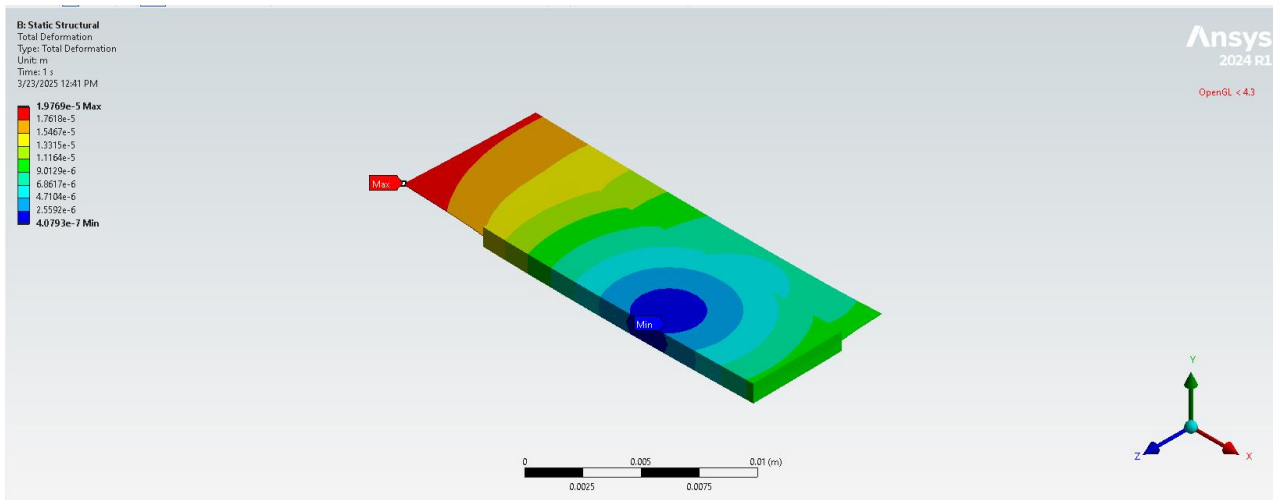


Fig 1.2: Total Deformation

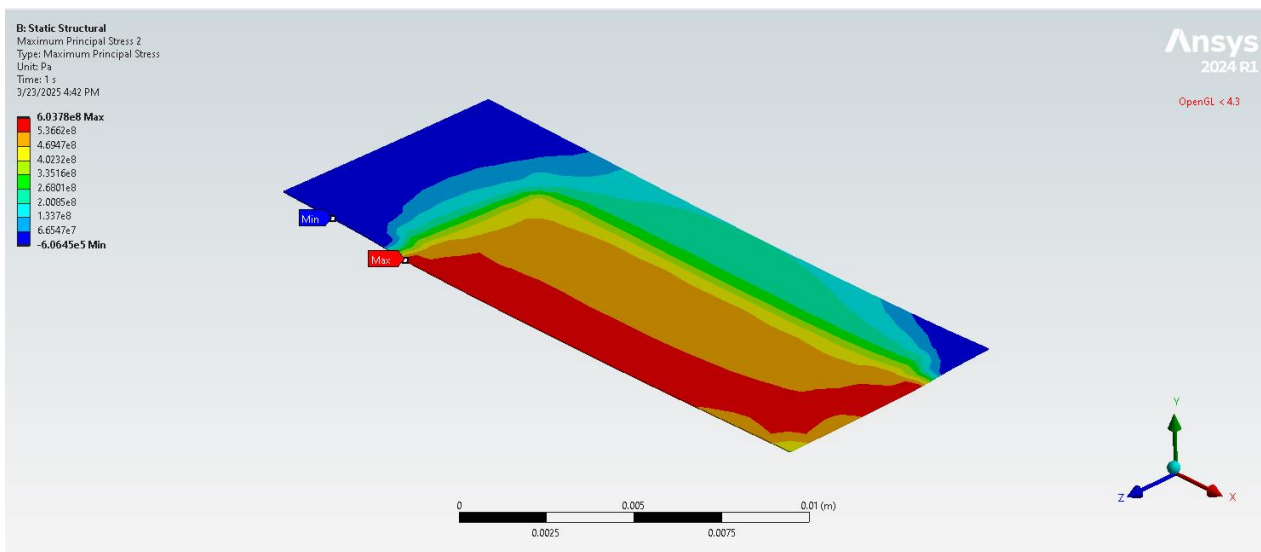


Fig 1.3: Maximum Principal Stress Across ARC Layer

As observed from the results, the deformation is observed across the entire structure. The maximum stress is around the ARC layer; concentrated around the centre of the plane. Hence, it is implied that the ARC layer cannot sustain through the temperature change with the given dimensions and porosity levels.

Results Interpretation:

In the simulator, the thicknesses t_{Cut} and t_{NCut} are defined as input parameters, along with the physical properties of Nano-Cu, which vary with porosity. For t_{NCut} and t_{NCu} values of 5 μm , 10 μm the copper width varies within a range of 1 Oz to 20 Oz. The maximum stress, equivalent thermal conductivity, and electrical resistivity are determined.

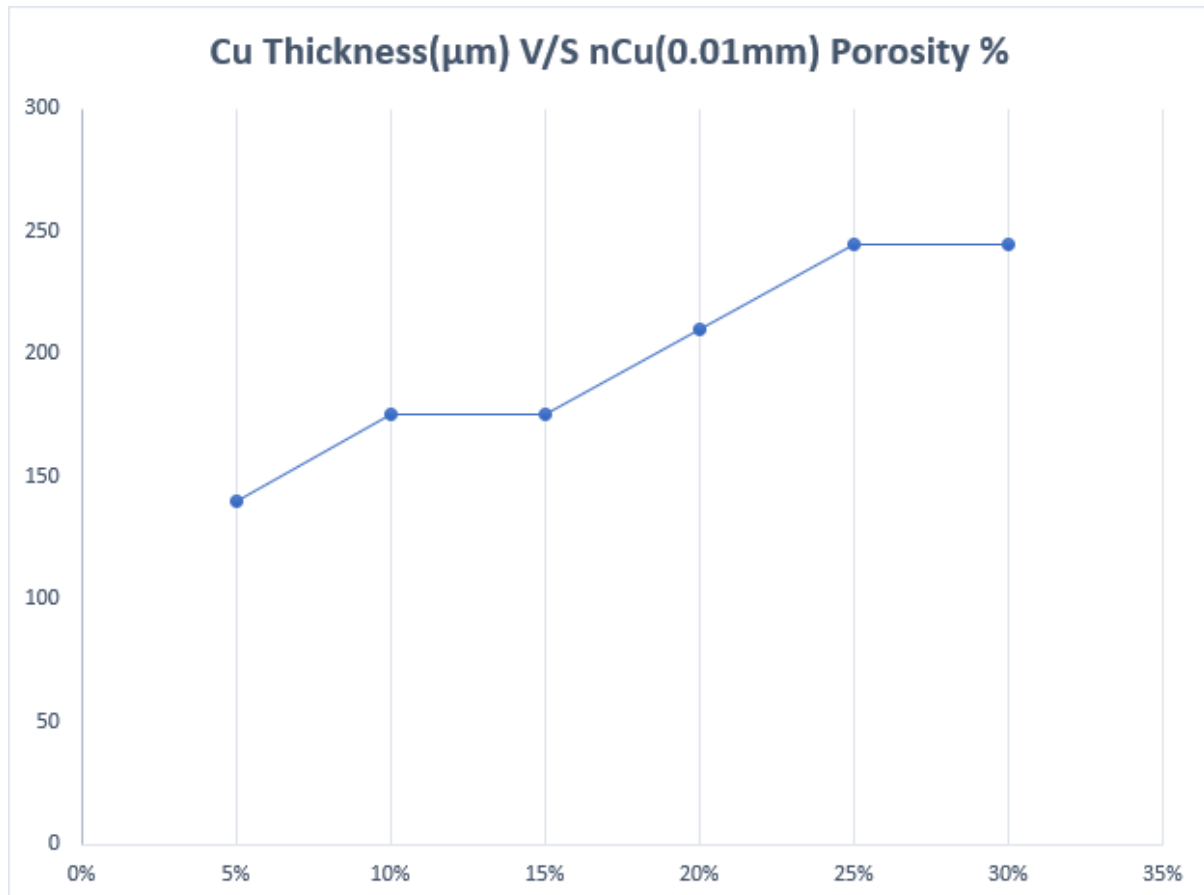


Fig 1.4: Copper Thickness Variations with Changes in Porosity 10 μm case

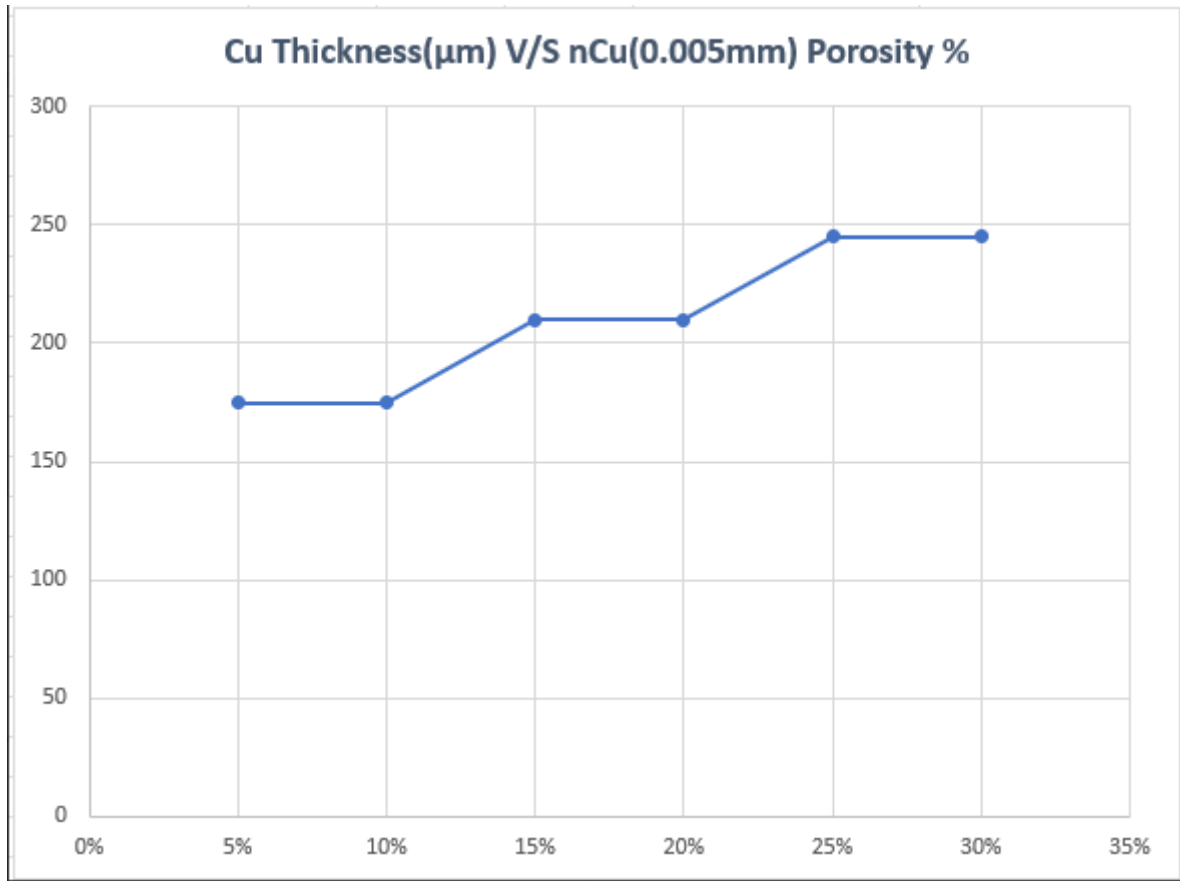


Fig 1.5: Copper Thickness Variations with Changes in Porosity 5μm case

When the t_{NCu} is 5 μm, the copper thickness is varied from 1 Oz (35 μm) to 20 Oz (690 μm). This variation significantly impacts the stress-strain profile. For instance, at 5% porosity, the target safe limit of 550 MPa is reached at a copper thickness of 175 μm. Additionally, for porosity levels of 5% and 30%, it was observed that the point where stress approaches the maximum limit of 600 MPa shifts lower on the graph. This indicates that the maximum allowable copper thickness decreases as porosity increases, thereby reducing the structure's stress tolerance. The optimum range of copper to be used for achieving the required stress profile is around 4 to 6 Oz.

Prob. 1.2:

Equivalent Electrical Resistivity:

The resistivity of copper changes with temperature. The temperature coefficient of resistivity for copper (α_{Cu}) is used to account for this change in resistivity as the temperature varies from the reference temperature (22°C) to the applied temperature (150°C). The formula used to adjust the resistivity of copper based on temperature is:

$$\rho_{Cu,T} = \rho_{Cu,20^\circ C} * (1 + \alpha_{Cu} * (T_{applied} - T_{ambient})) \quad [2]$$

Where,

- $\rho_{Cu,T}$: resistivity of copper at the applied temperature $T_{applied}$,
- $\rho_{Cu,20^\circ C}$: resistivity of copper at $20^\circ C$,
- α_{Cu} : temperature coefficient of resistivity for copper,
- $T_{applied}$: applied temperature,
- $T_{ambient}$: ambient temperature.

Hence, we adjust the resistivity of copper based on the operating temperature. As temperature increases, the resistivity of copper increases due to increased atomic vibrations, which hinder electron flow.

Nano-Cu is modelled as a material whose resistivity is influenced by its porosity. The porosity in nano-copper increases its resistivity. The formula used to scale the resistivity of nano-copper based on porosity is:

$$\rho_{NanoCu} = \rho_{Cu} * (1 - \text{porosity}/100)^{-n}$$

Where,

- ρ_{NanoCu} : resistivity of nano-copper at a given porosity,
- ρ_{Cu} : resistivity of copper (adjusted for temperature),
- porosity is the porosity level in percentage,
- n : an empirical exponent that accounts for the non-linear effect of porosity on resistivity.

Porosity introduces voids in the material, reducing its effective cross-sectional area for conduction. The scaling factor models the increase in resistivity with increasing porosity.

The per-square resistance is computed using the resistivities of copper and nano-copper, considering their respective thicknesses. The formula for calculating the per-square resistance for a composite material made up of copper and nano-copper is:

$$R_{square} = 1 / ((t_{Cu} / \rho_{Cu}) + (t_{NanoCu} / \rho_{NanoCu}))$$

Where:

- - t_{Cu} and t_{NanoCu} : thicknesses of copper and nano-copper, respectively,

This formula reflects that the equivalent resistance is a parallel contribution of each material e , given that they are treated as being in parallel for current conduction. During this analysis, standard variations in porosity and Cu and NanoCu thickness levels are assumed as done previously.

Fringing effects, which result from the non-uniform electric field near material edges, are ignored in this methodology as their impact is minimal for the given conditions. This simplification is justified since the methodology we use primarily deals with very small thicknesses ($5 \mu m$ and $10 \mu m$), where edge effects are negligible compared to the bulk material. Additionally, the large surface area assumption ensures that peripheral edge effects have an insignificant influence on overall resistance. We also assume a linear relationship between resistivity, temperature, and porosity, disregarding any potential non-linear behaviour at the edges. Furthermore, per-square resistance is largely determined by the bulk properties of the material, making edge effects inconsequential for typical use cases.

A MATLAB code was developed for the same and the resistivity variations were plotted.

--- Results for $t_{\text{NCu}} = 5.0 \mu\text{m}$ at $T = 150^\circ\text{C}$ ---

Equivalent Per-Square Resistance (Ohms/sq):

	P0%	P5%	P10%	P15%	P20%	P25%	P30%
tCu=1oz	0.00071502	0.00071576	0.00071649	0.0007172	0.00071789	0.00071856	0.00071921
tCu=2oz	0.00036004	0.00036023	0.00036042	0.0003606	0.00036077	0.00036094	0.00036111
tCu=3oz	0.0002406	0.00024068	0.00024077	0.00024085	0.00024092	0.000241	0.00024107
tCu=4oz	0.00018066	0.00018071	0.00018076	0.0001808	0.00018085	0.00018089	0.00018093
tCu=5oz	0.00014463	0.00014466	0.00014469	0.00014472	0.00014475	0.00014478	0.0001448
tCu=6oz	0.00012058	0.00012061	0.00012063	0.00012065	0.00012067	0.00012069	0.0001207
tCu=7oz	0.00010339	0.00010341	0.00010342	0.00010344	0.00010345	0.00010347	0.00010348
tCu=8oz	9.0492e-05	9.0504e-05	9.0516e-05	9.0527e-05	9.0538e-05	9.0549e-05	9.0559e-05
tCu=9oz	8.0454e-05	8.0463e-05	8.0472e-05	8.0481e-05	8.049e-05	8.0498e-05	8.0506e-05
tCu=10oz	7.242e-05	7.2427e-05	7.2435e-05	7.2442e-05	7.2449e-05	7.2456e-05	7.2463e-05
tCu=11oz	6.5845e-05	6.5851e-05	6.5857e-05	6.5863e-05	6.5869e-05	6.5875e-05	6.588e-05
tCu=12oz	6.0364e-05	6.0369e-05	6.0375e-05	6.038e-05	6.0385e-05	6.0389e-05	6.0394e-05
tCu=13oz	5.5726e-05	5.573e-05	5.5735e-05	5.5739e-05	5.5743e-05	5.5747e-05	5.5751e-05
tCu=14oz	5.1749e-05	5.1753e-05	5.1757e-05	5.1761e-05	5.1764e-05	5.1768e-05	5.1771e-05
tCu=15oz	4.8303e-05	4.8306e-05	4.831e-05	4.8313e-05	4.8316e-05	4.8319e-05	4.8322e-05
tCu=16oz	4.5287e-05	4.529e-05	4.5292e-05	4.5295e-05	4.5298e-05	4.5301e-05	4.5303e-05
tCu=17oz	4.2625e-05	4.2628e-05	4.263e-05	4.2633e-05	4.2635e-05	4.2637e-05	4.264e-05
tCu=18oz	4.0259e-05	4.0261e-05	4.0263e-05	4.0266e-05	4.0268e-05	4.027e-05	4.0272e-05
tCu=19oz	3.8141e-05	3.8144e-05	3.8146e-05	3.8148e-05	3.815e-05	3.8151e-05	3.8153e-05
tCu=20oz	3.6236e-05	3.6238e-05	3.6239e-05	3.6241e-05	3.6243e-05	3.6245e-05	3.6246e-05

Fig. 1.6 Equivalent Electrical Resistivity Approximation for $t_{\text{NCu}} = 5\mu\text{m}$

--- Results for $t_{\text{NCu}} = 10.0 \mu\text{m}$ at $T = 150^\circ\text{C}$ ---

Equivalent Per-Square Resistance (Ohms/sq):

	P0%	P5%	P10%	P15%	P20%	P25%	P30%
tCu=1oz	0.00070509	0.00070654	0.00070796	0.00070935	0.0007107	0.00071202	0.0007133
tCu=2oz	0.00035751	0.00035788	0.00035825	0.0003586	0.00035895	0.00035928	0.00035961
tCu=3oz	0.00023946	0.00023963	0.00023979	0.00023995	0.00024011	0.00024026	0.0002404
tCu=4oz	0.00018002	0.00018012	0.00018021	0.0001803	0.00018039	0.00018047	0.00018055
tCu=5oz	0.00014422	0.00014428	0.00014434	0.0001444	0.00014446	0.00014451	0.00014456
tCu=6oz	0.0001203	0.00012034	0.00012038	0.00012042	0.00012046	0.0001205	0.00012054
tCu=7oz	0.00010318	0.00010321	0.00010324	0.00010327	0.0001033	0.00010333	0.00010336
tCu=8oz	9.0331e-05	9.0355e-05	9.0378e-05	9.0401e-05	9.0423e-05	9.0444e-05	9.0465e-05
tCu=9oz	8.0326e-05	8.0345e-05	8.0364e-05	8.0381e-05	8.0399e-05	8.0416e-05	8.0432e-05
tCu=10oz	7.2317e-05	7.2332e-05	7.2347e-05	7.2361e-05	7.2375e-05	7.2389e-05	7.2402e-05
tCu=11oz	6.5759e-05	6.5772e-05	6.5784e-05	6.5796e-05	6.5808e-05	6.5819e-05	6.583e-05
tCu=12oz	6.0292e-05	6.0303e-05	6.0313e-05	6.0323e-05	6.0333e-05	6.0343e-05	6.0352e-05
tCu=13oz	5.5665e-05	5.5674e-05	5.5683e-05	5.5691e-05	5.5699e-05	5.5708e-05	5.5715e-05
tCu=14oz	5.1697e-05	5.1705e-05	5.1712e-05	5.172e-05	5.1727e-05	5.1734e-05	5.174e-05
tCu=15oz	4.8257e-05	4.8264e-05	4.827e-05	4.8277e-05	4.8283e-05	4.8289e-05	4.8295e-05
tCu=16oz	4.5246e-05	4.5252e-05	4.5258e-05	4.5264e-05	4.5269e-05	4.5274e-05	4.528e-05
tCu=17oz	4.2589e-05	4.2594e-05	4.26e-05	4.2605e-05	4.2609e-05	4.2614e-05	4.2619e-05
tCu=18oz	4.0227e-05	4.0232e-05	4.0236e-05	4.0241e-05	4.0245e-05	4.0249e-05	4.0253e-05
tCu=19oz	3.8113e-05	3.8117e-05	3.8121e-05	3.8125e-05	3.8129e-05	3.8133e-05	3.8137e-05
tCu=20oz	3.621e-05	3.6214e-05	3.6217e-05	3.6221e-05	3.6225e-05	3.6228e-05	3.6231e-05

Fig. 1.7 Equivalent Electrical Resistivity Approximation for $t_{\text{NCu}} = 10\mu\text{m}$

Equivalent Thermal Conductivity:

This problem is sorted by considering the physical aspects of the structure. Since this is a quarter of the actual symmetry, it being asymmetrical, we have taken a linear approach for the calculation of R_{Th} . The area is found by multiplying the $W \cdot t$ for each layer and then multiplying by 4. The standard values of ‘k’ scale in accordance with the porosity. These are used as per the formula in the class slides of lecture #3 to find the respective thermal resistances R_{Th} .

ARC	nCu	tCu	P	Rth
mm	mm	mm	%	k/W
0.04	0.005	0.245	30%	0.004272
0.04	0.005	0.245	25%	0.004264
0.04	0.005	0.21	20%	0.003966
0.04	0.005	0.21	15%	0.00396
0.04	0.005	0.175	10%	0.003663
0.04	0.005	0.175	5%	0.003659

Table 1.1: R_{Th} calculation for $TNCu = 5\mu m$

ARC	nCu	tCu	P	Rth
mm	mm	mm	%	k/W
0.04	0.01	245	30%	0.00439
0.04	0.01	245	25%	0.004375
0.04	0.01	210	20%	0.00407
0.04	0.01	175	15%	0.003767
0.04	0.01	175	10%	0.003756
0.04	0.01	140	5%	0.003455

Table 1.2: R_{Th} calculation for $TNCu = 10\mu m$

Challenges and Mitigations:

Meshing was time-consuming due to the complex geometry, so a quarter-symmetry approach was used to reduce computation time while maintaining accuracy. This required careful boundary condition validation to ensure reliable results.

Problem 2: Half Bridge power Module for a Synchronous Buck Converter

Problem Statement:

This task details the design, simulation, and layout of a $\frac{1}{2}$ Bridge Power Module (HBPM) for a synchronous buck converter. The electrical design was simulated at an input voltage of **120 V** and output voltage of **12 V**, using Wolf-speed 1200V/25m Ω SiC MOSFETs. Key parameters such as switching characteristics, output ripple, and efficiency were analysed.

The physical layout was developed on a 635 μm Al₂O₃ DBC substrate, with MOSFETs soldered to a 2 mm copper baseplate. Wire bonds and solder interconnects were used following DBC design guidelines. A 3D model was created in SolidWorks.

Thermal analysis included determining copper interconnect thicknesses and junction-to-case thermal resistance $R_{\theta jc}$, ensuring a die temperature of 70°C ambient. A 2D thermal model was developed to optimize heat dissipation.

Design and analysis of synchronous buck converter:

We considered the following buck converter system for our half bridge converter:

V_{in}	120 V
V_{out}	12 V
Load 100%	360 W
Load 10%	36 W
$V_{ripple\ pk-pk}\ (10\%)$	1.2 V
$I_{ripple\ pk-pk}\ (10\%)$	3 A

Table 1.1: Buck Converter Design Parameters

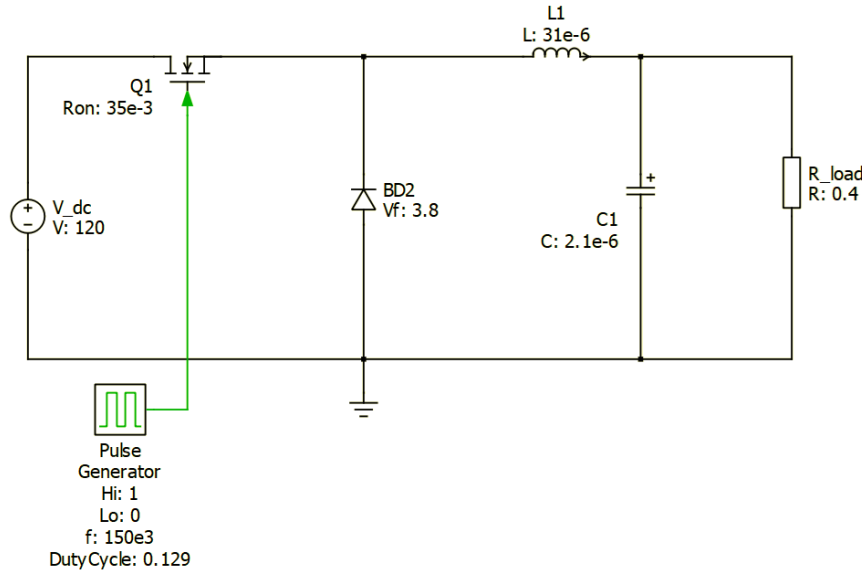


Figure 2.1: Buck Converter Model with Q2 Shown as Freewheeling Body Diode

Figure 2.1 shows the ideal model of the buck converter. Switch 2 has been replaced with a diode in the diagram to indicate the body diode as the freewheeling current path. Diode voltage drops and drain-source resistances were included in the design of the buck converter. Nonidealities in the inductor and capacitor were ignored.

Small ripple approximation was performed for each state.

When Q1 is ON: $v_L = -i * R_{ds,on} + V_{in} - v_o$ $i_c = i - \frac{v_o}{R}$

When Q1 is OFF: $v_L = -V_f - v_o$ $i_c = i - \frac{v_o}{R}$

Volt-second balance: $\langle v_L \rangle = D(V_{in} - I * R_{ds,on}) - D'V_f - V_o = 0$

Charge-Second Balance: $\langle i_c \rangle = D \left(I - \frac{V_o}{R} \right) + D' \left(I - \frac{V_o}{R} \right) = 0$

Given these equations and our design parameters, we can solve for $D = 0.129$

The design guide [1] was followed for the selection of the inductor and capacitor values. Given our design parameters for ripples, we can state that:

$$L = \frac{V_{in} - V_o - I R_{ds,on} D}{\Delta i_L * f_{sw}} * D = 30.92 \mu H \quad C = \frac{\Delta i_L}{8 * f_{sw} * \Delta v_o} = 2.1 \mu F$$

From the MOSFET datasheet [2], an estimate for Drain source Resistance at 70° C was made. This is used for loss calculations.

$$R_{case} = 70^\circ C \quad R_{th,jc} = 0.25^\circ C / W \quad I_{load,max} = 30A \quad R_{ds,on} (first\ guess) = 35m\Omega$$

$$T_j = T_{case} + R_{th,jc} * I_{load,max}^2 * R_{ds,on} \quad T_j = 78^\circ C \quad R_{ds,on} = 35m\Omega$$

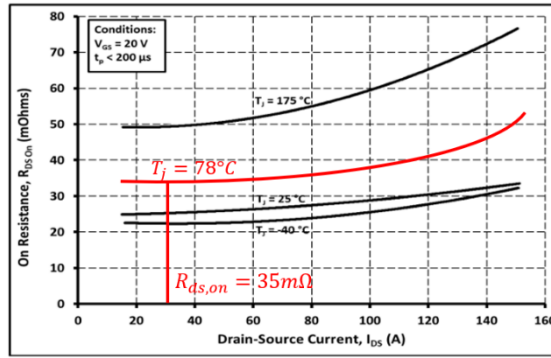


Figure 5.

On-Resistance vs. Drain Current For Various Temperatures

Figure 1.1 SiC MOSFET $R_{ds,on}$ for varying Junction Temp and Load

Simulation in LTSpice

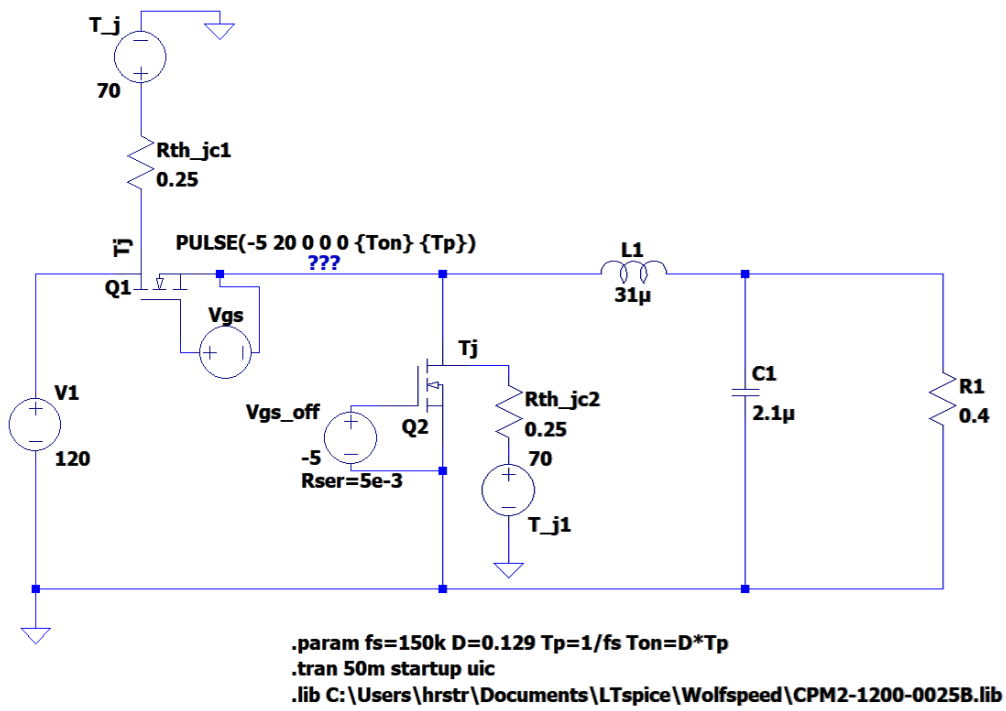


Figure 2.2 LT Spice Simulation Model

A model, shown in figure 2.2, was developed in LTSpice to simulate the performance of the synchronous buck converter. L and C were treated as ideal and the Wolf-Speed provided MOSFET module was used. A thermal network was added to set the junction temperature based on an assumed case temperature of 70°C.

Full Load Case:

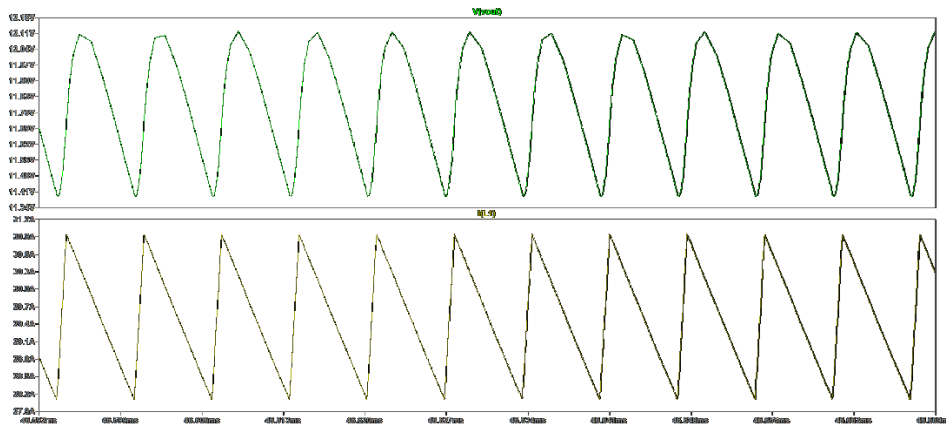


Figure 3.3 Full Load Inductor and Current Ripples

The operation at full load is taken from the plots as follows: $dV = 0.7V$, $dI = 2.8A$, $V_o = 11.8V$, $I_o = 29.5A$

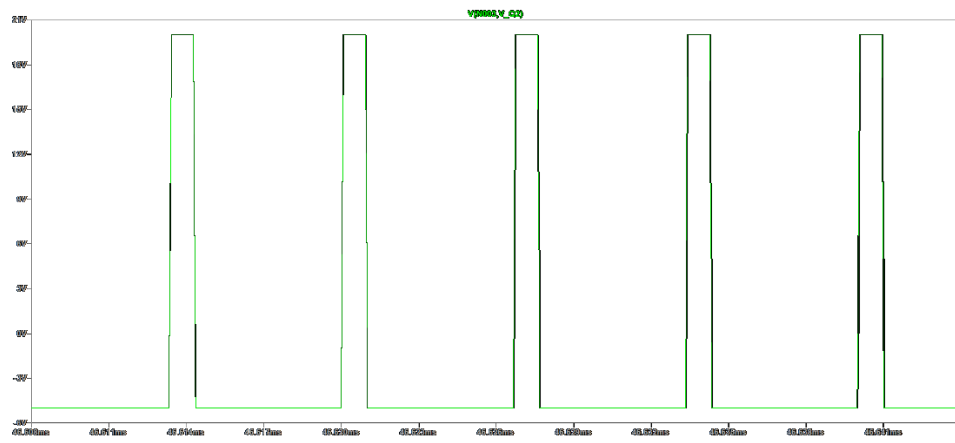


Figure 2.44 Gate Signal for Q1. D=0.129

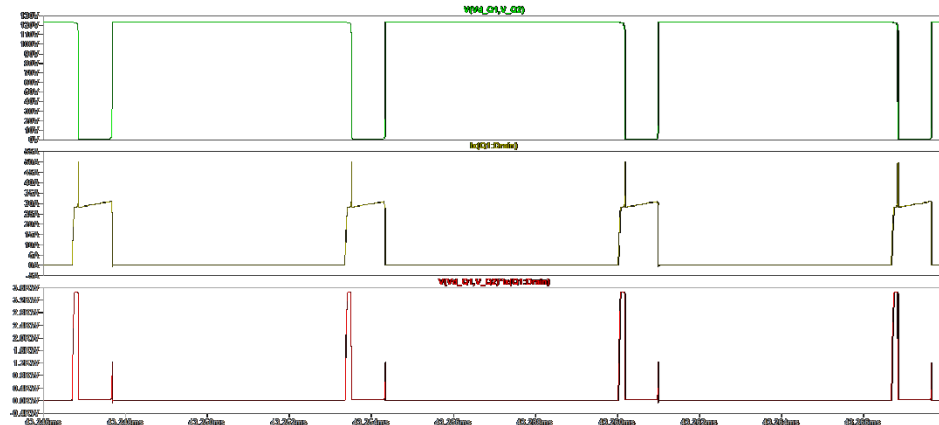


Figure 2.5 Full Load Switching Waveforms Q1 (top: V_{ds} [V], center: I_{ds} [A], bottom: Power Loss [W])

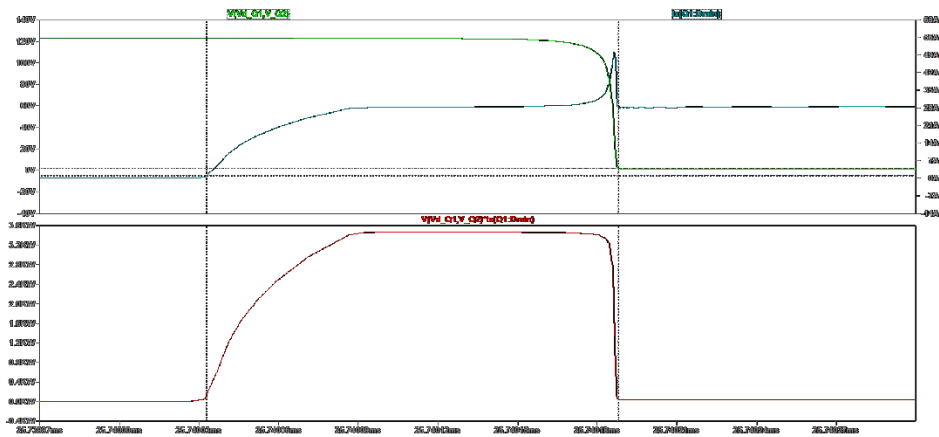


Fig. 2.6: Q1 Turn-On Loss Calculations

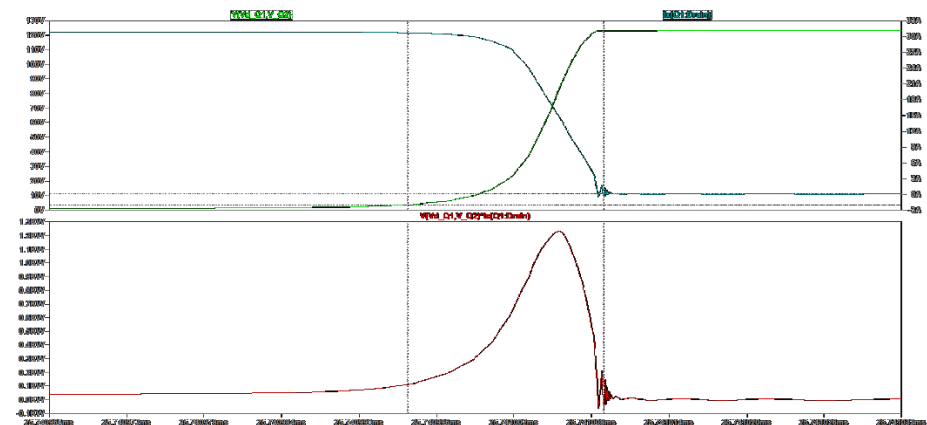


Fig. 2.6.1: Q1 Turn-Off Loss Calculations

Q1 Detailed power loss calculations:

$$P_{turnon} = E_{sw,on} * f_{sw} = 465.39\mu J * 150kHz = 69.81 W$$

$$P_{turnoff} = E_{sw,off} * f_{sw} = 8.53\mu J * 150kHz = 1.28 W$$

$$P_{conduction} = I_d^2 * R_{ds,on} * D = 30^2 * 0.129 = 4.06 W$$

$$P_{Q2} = P_{turn,on} + P_{turn,off} + P_{conduction} = 69.81 + 0.28 + 4.06 = 74.15 W$$

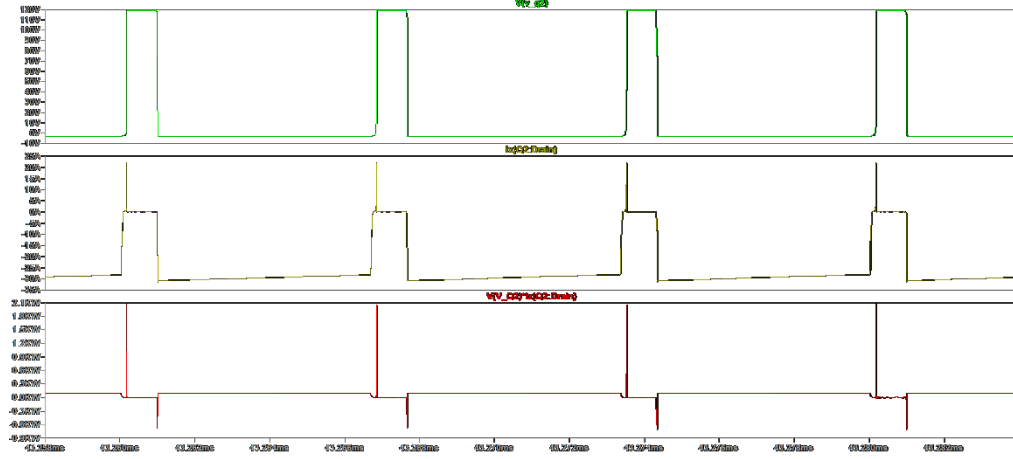


Figure 2.5 Full Load Switching Waveforms Q2 (top: Vds [V], center: Ids [A], bottom: Power Loss [W])

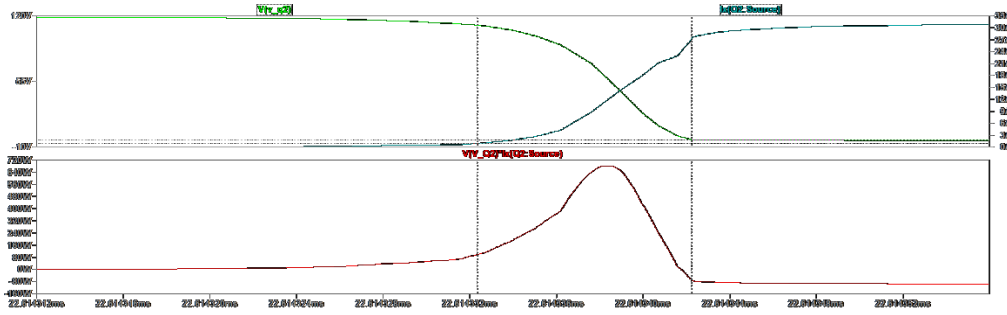


Fig 2.7.1 Q2 Turn on Loss Calculation

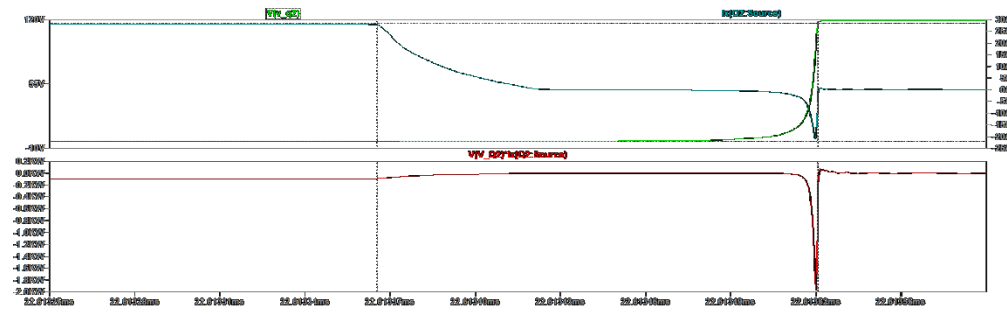


Fig 2.7.2 Q2 Turn off Loss Calculation

Q2 Detailed power loss calculations:

$$P_{turnon} = E_{sw,on} * f_{sw} = 3.47\mu J * 150kHz = 0.52 W$$

$$P_{turn,off} = E_{sw,off} * f_{sw} = 5.41\mu J * 150kHz = 0.81 W$$

$$P_{conduction} = V_f * I_d * D' = 3.8 * 30 * (1 - 0.129) = 99.29 W$$

$$P_{Q2} = P_{turn,on} + P_{turn,off} + P_{conduction} = 99.29 + 0.81 + 0.52 = 100.63 W$$

Junction Temperature Estimates from Wolf-Speed LTSpice Model: Q1: 89.58°C, Q2: 90.68°C

10% Load Case:

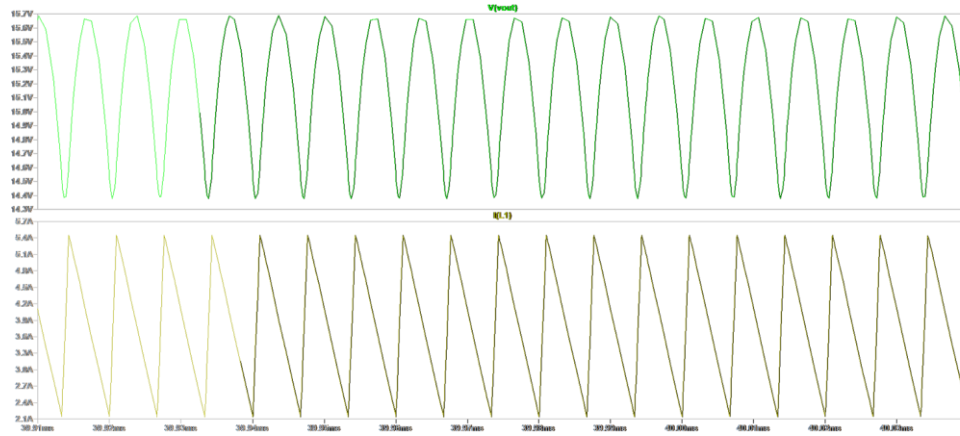


Figure 2.6 10% Load Inductor and Current Ripples

The operation at full load is taken from the plots as follows: $dV = 1.3V$, $dI = 3.3A$, $V_o = 15.2V$, $I_o = 3.8A$

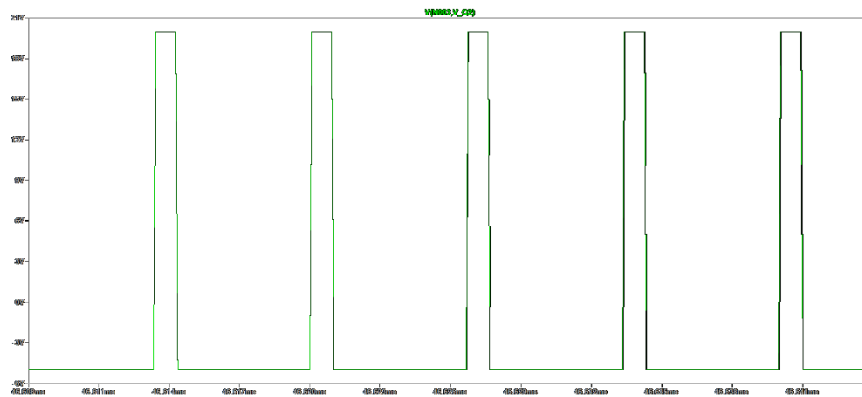


Figure 2.7 Gate Signal for Q1. $D=0.129$

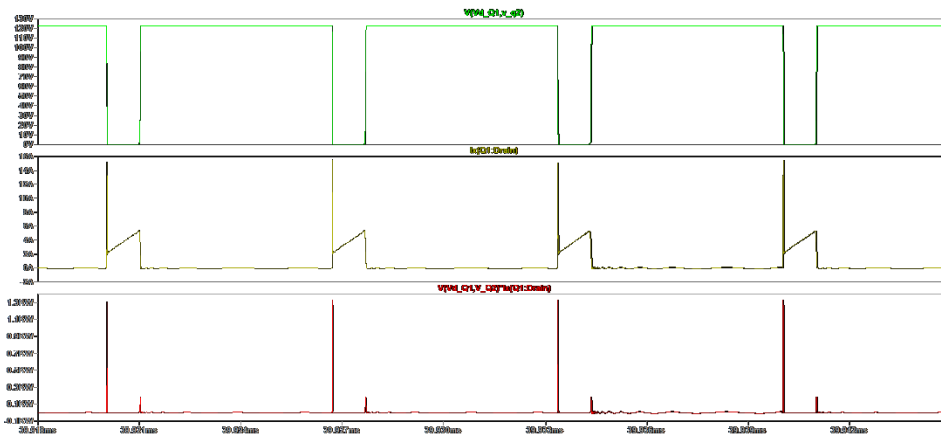


Figure 2.8 10% Load Switching Waveforms Q1 (top: V_{ds} [V], center: I_{ds} [A], bottom: Power Loss [W])

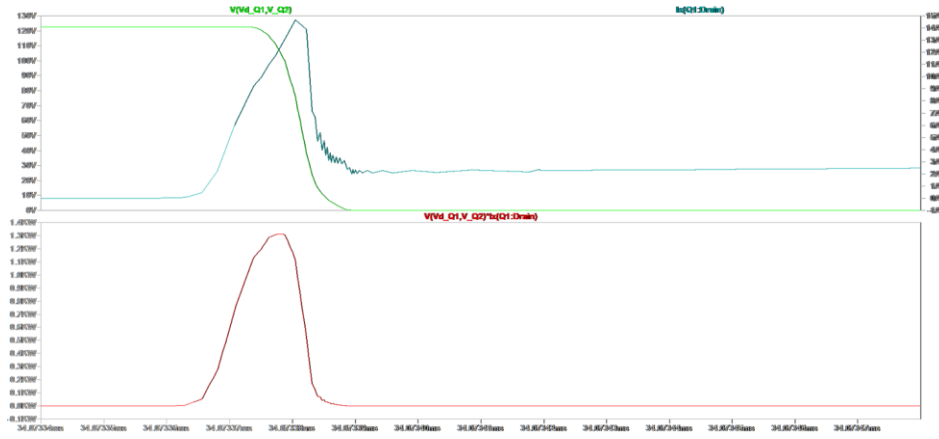


Figure 2.9 Q1 Turn on Loss Calculations

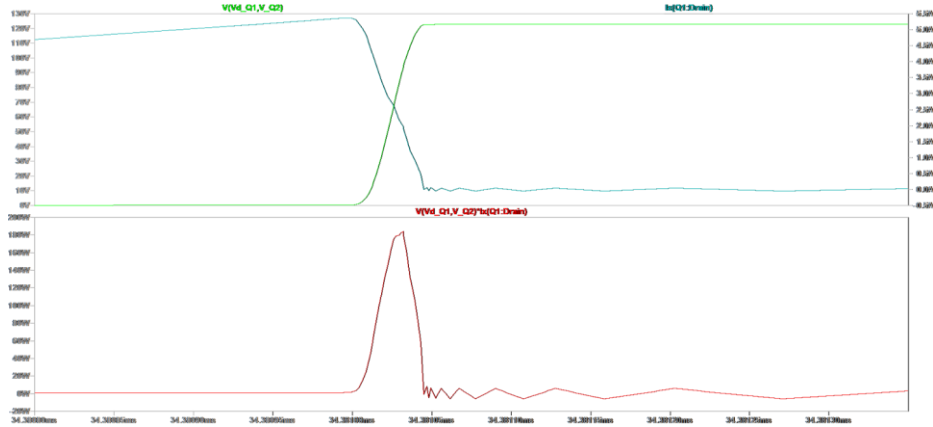


Figure 2.12 Q1 Turn off Loss Calculations

Q1 Detailed power loss calculations:

$$P_{turnon} = E_{sw,on} * f_{sw} = 10.027\mu J * 150kHz = 1.50 W$$

$$P_{turn,off} = E_{sw,off} * f_{sw} = 4.34\mu J * 150kHz = 0.65 W$$

$$P_{conduction} = I_d^2 * R_{ds,on} = 3^2 * 0.035 = 0.32 W$$

$$P_{Q1} = P_{turn,on} + P_{turn,off} + P_{conduction} = 1.5 + 0.65 + 0.32 = 2.47 W$$

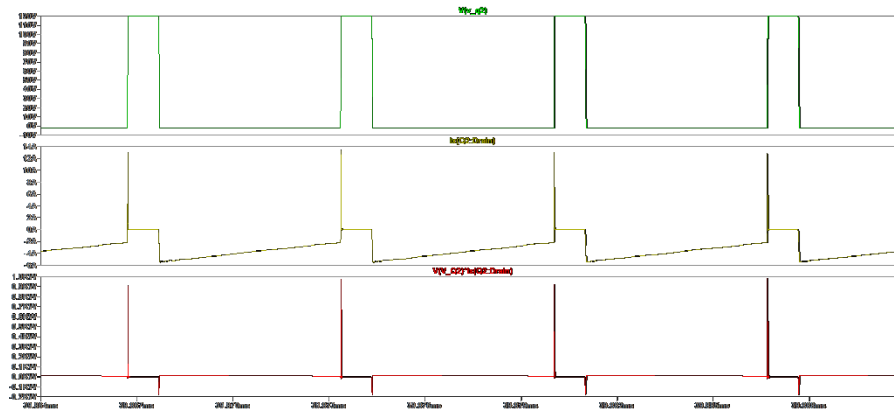


Figure 2.13 10% Load Switching Waveforms Q2 (top: V_{ds} [V], center: I_{ds} [A], bottom: Power Loss [W])

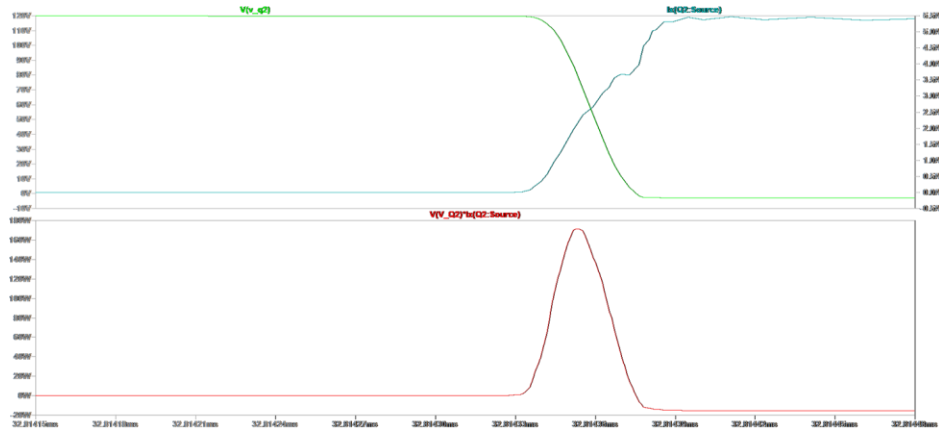


Figure 2.14 Q2 Turn on Loss Calculations

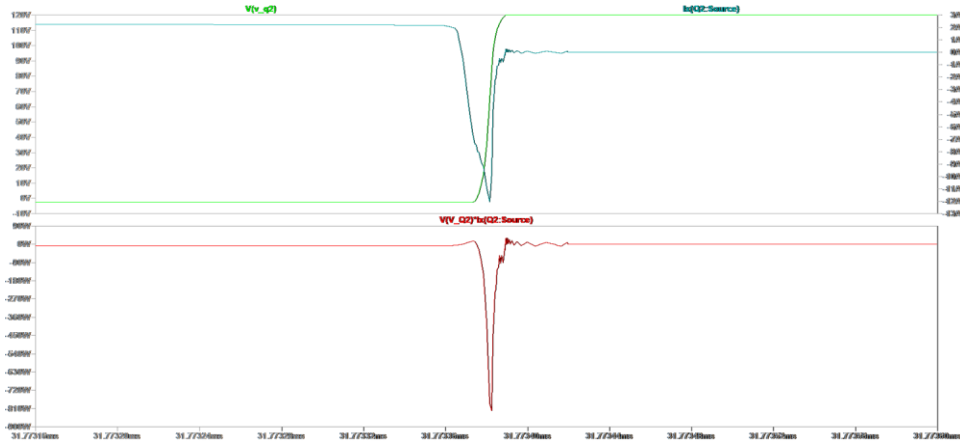


Figure 2.15 Q2 Turn off Loss Calculations

Q2 Detailed power loss calculations:

$$P_{turnon} = E_{sw,on} * f_{sw} = 3.80\mu J * 150kHz = 0.57 W$$

$$P_{turn,off} = E_{sw,off} * f_{sw} = 3.48\mu J * 150kHz = 0.52 W$$

$$P_{conduction} = V_f * I_d * D' = 3.8 * 3.8 * (1 - 0.129) = 12.58 W$$

$$P_{Q2} = P_{turn,on} + P_{turn,off} + P_{conduction} = 0.57 + 0.52 + 12.58 = 13.67 W$$

Junction Temperature Estimates from Wolf-Speed LTSpice Model: Q1: 72.15°C, Q2: 72.57°C

Half-Bridge Converter Parameters and History

Sr. No	Parameters	Values
1	V_{in}	120 V
2	V_{out}	12 V
3	Load 100%	360 W
4	Load 10%	36 W
5	Switching Frequency	150 kHz
6	Inductor	30.92 μ H
7	Capacitor	2.1 μ F
8	Duty Cycle	0.129
9	$V_{ripple\ pk-pk}$ (10%) at 100% load	1.2 V
10	$I_{ripple\ pk-pk}$ (10%) at 100% load	3 A
11	P_{Q1} at 100% load	74.15 W
12	P_{Q2} at 100% load	100.63 W
13	P_{Q1} at 10% load	2.47 W
14	P_{Q2} at 10% load	13.67 W
15	Measured $V_{ripple\ pk-pk}$ at 100% load	0.7 V
16	Measured $I_{ripple\ pk-pk}$ at 100% load	2.8 A
17	Measured $V_{ripple\ pk-pk}$ at 10% load	1.3 V
18	Measured $I_{ripple\ pk-pk}$ at 10% load	3.3 A
19	V_{out} at 100% load	11.8 V
20	I_{out} at 100% load	29.5 A
21	V_{out} at 10% load	15.2 V
22	I_{out} at 10% load	3.8 A

One takeaway from the results is the poor performance in the 10% loading cases. The output voltage and current is not as targeted. There are two primary drivers of this discrepancy. The first reason is the reduced voltage drop due to drain-source resistance at low loads. The other reason is the reduced voltage drop across the body diode, as we are operating in the unsaturated region. A control circuit to modify duty cycle would resolve these errors.

Physical Layout: Half Bridge Power Module

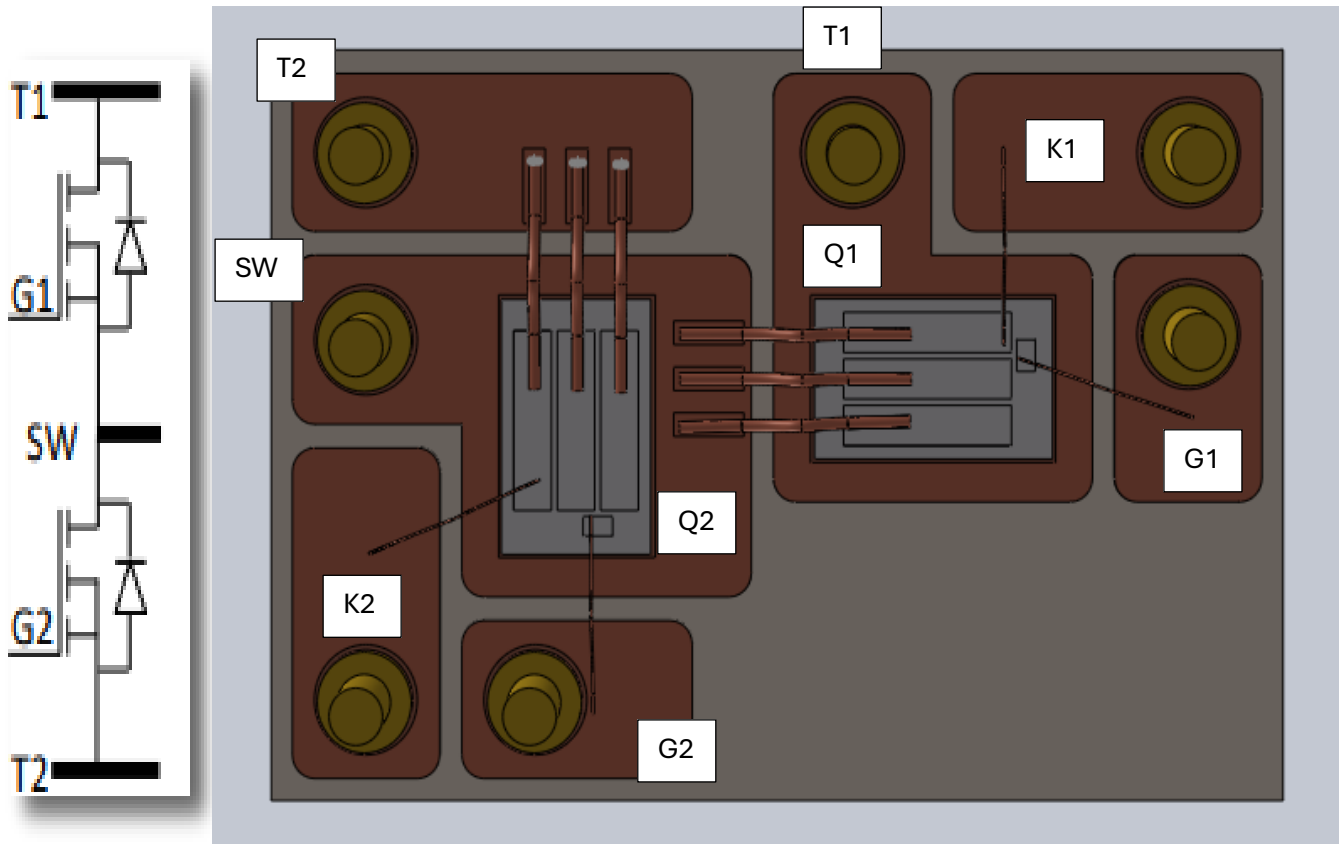


Figure 2.16 Layout

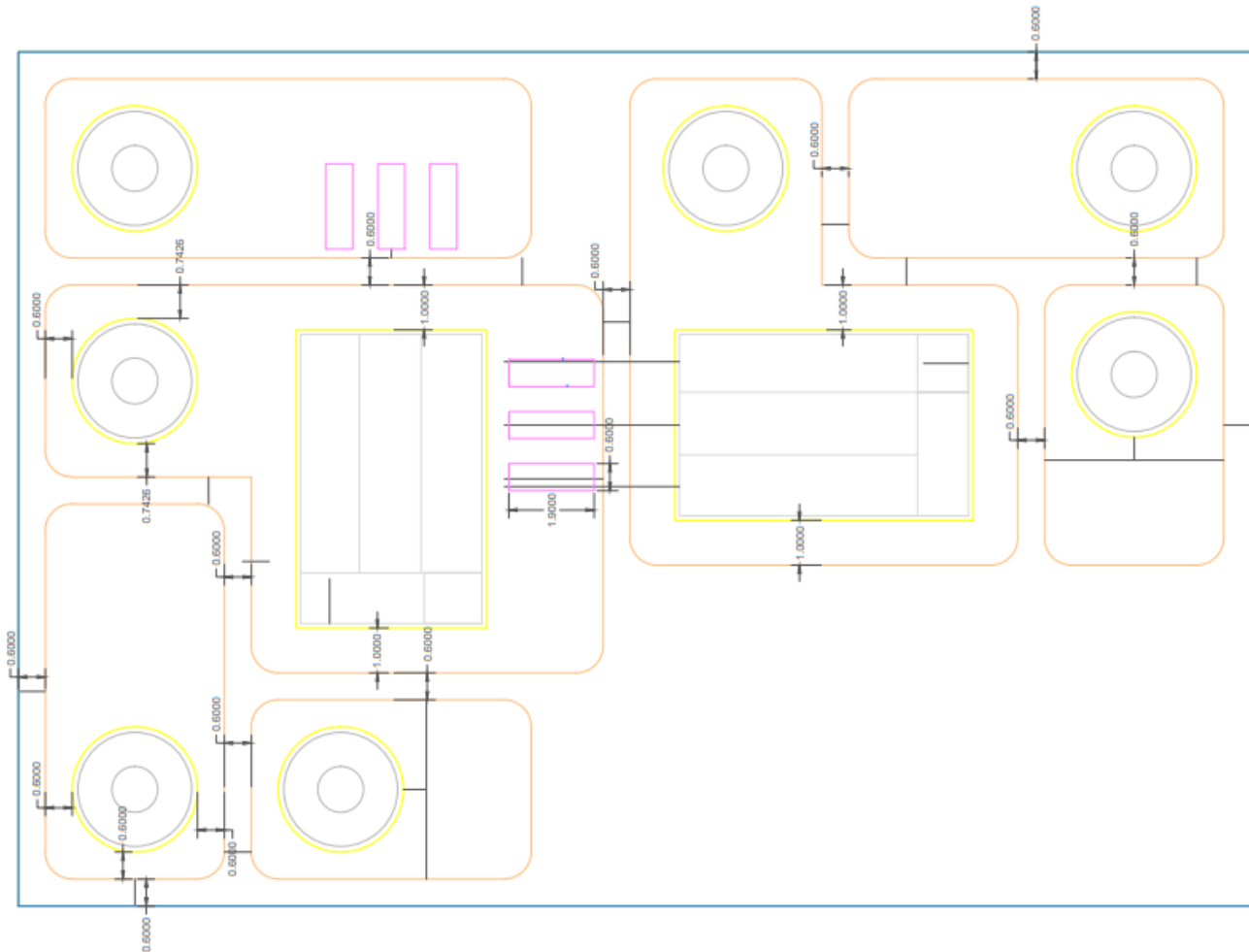
This is the layout for half bridge power model. Q1&Q2 are MOSFETs, G1&G2 are gate terminals, K1&K2 are kelvin terminals, T1 is Drain for MOSFET 1, T2 is Source for MOSFET 2, SW terminal that connects to Inductor. All terminals are labelled in the figure above. The SiC MOSFET die dimensions for CPM2-1200-0025B is available in [3]. The Dimensions of each layer in Power Module are listed in Table-2.

Table 2 Physical Dimensions of each layer in Power Module

Parameters	Layer	Materials	Thickness (mm)
Module layout (27.5mm×19mm)	Die	SiC	0.18
	DBC	Copper	0.25
	Substrate	Alumina	0.635
	Solder	SAC305	0.05
	Copper Baseplate	Copper	2

Distances and Spacing:

The following figure shows the distances and spaces for the Half Bridge Power Module



For the distances and spacing DBC design guide is followed. The copper thickness is selected 10 mils (0.25mm) which is smaller than ceramic thickness. The line widths are more than required, in this case the line widths are equal to or more than 4.28mm. This distance accommodates terminals bottom diameter and minimum 20mils of distance between terminal and edge of DBC. The radius of curvature is taken as 20 mils which is in the limits of 20 mils – 60 mils

Source Wire size: Three wires in parallel. Nominal Current = $30/3 = 10\text{A}$. Wire size shall be designed for 3x the current rating to handle any overload conditions, so each wire should be rated for at least 30A.

From the table below: 300 μm diameter, no more than 7.5mm length

Gate Wire size:

From LTSpice: Gate current RMS $\approx 300\text{mA}$. Increasing to 2.4A for added robustness. 50 μm diameter, 5mm length

The Following table is followed for the wire size

Wire Dia μm (mils)	A1 (99.99%)						Al-1% Si					
	Wire length (mm)						Wire length (mm)					
	1.3	2.5	5	7.5	10	12.5	1.3	2.5	5	7.5	10	12.5
25 (1)	1.2	0.6	0.3	0.2	0.2	0.1	0.8	0.4	0.2	0.1	0.1	0.1
50 (2)	4.8	2.4	1.2	0.8	0.6	0.5	3.2	1.6	0.8	0.5	0.4	0.3
75 (3)	11	5.4	2.7	1.8	1.4	1.1	7.2	3.6	1.8	1.2	0.9	0.7
100 (4)	19	9.6	4.8	3.2	2.4	1.9	13	6.4	3.2	2.1	1.6	1.3
125 (5)	30	15	7.5	5.0	3.8	3.0	20	10	5.0	3.3	2.5	2.0
200 (8)		38	19	13	9.6	7.7		26	13	8.5	6.4	5.1
250 (10)		60	30	20	15	12		40	20	13	10	8.0
300 (12)		86	43	29	22	17		58	29	19	14	12
375 (15)		135	68	45	34	27			45	30	23	18
450 (18)		194	97	65	49	39			65	43	32	26
500 (20)		240	120	80	60	48			80	53	40	32
625 (25)		375	188	125	94	75			125	83	63	50

Figure 2.17 Bond Wire Fusing Currents

Pad Widths of Wolf-Speed MOSFET are large enough for the 12mil wire. The gate pad width is similarly large enough for 2 mil wire (not shown in chart)

Maximum-Useful Bond Pad dimensions					
Wire Size		Pad Width (W)		Pad Length (L+T)	
mil	μm	mil	μm	mil	μm
4	100	7.7	200	24	610
5	125	9.3	240	29	750
6	150	11	280	33	840
7	175	13	320	40	1010
8	200	14	370	44	1130
10	250	19	490	56	1410
12	300	23	580	67	1710
14	350	27	680	77	1960
15	375	28	720	83	2110
16	400	30	770	88	2240
18	450	34	860	99	2520
20	500	38	960	110	2800

Figure 2.18 Bond Wire Bad Dimensions

Following terminal is used in the model: Terminal: 1440-0-00-15-00-00-03-0

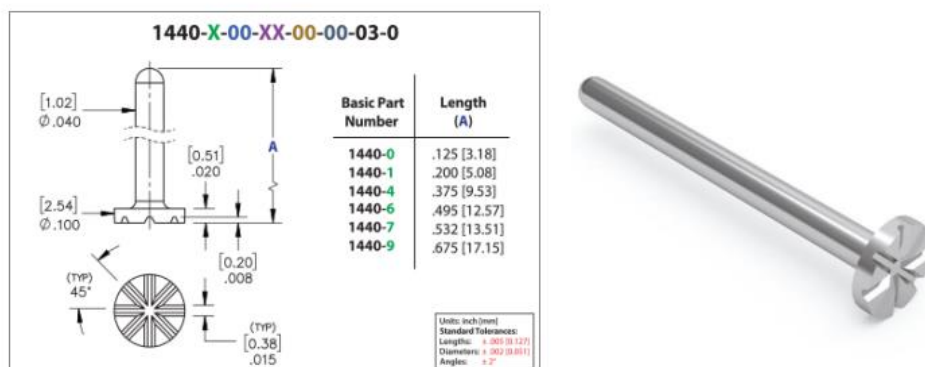


Figure 2.19 Terminal Specifications

Freewheeling Diode Path

The aim of this problem is to determine the copper interconnect thicknesses and line widths for the “freewheeling current path” by keeping the resistance to $\leq 3\%$ of the $R_{DS(on)}$ of the MOSFET for your DBC Layout. The resistance is measured between terminal tips.

The Current free-wheeling part is labelled as 1-2-3. Hence the current enters through terminal T2, then goes MOSFET 2 and through its body diode it flows towards SW and from that terminal current flows towards inductor. So, the path resistance consists of both terminal resistances and resistance of both DBC. The analysis is as follows,

R1: - Terminal 1 resistance

$$A = 0.817mm^2$$

$$l = 3.18mm$$

$$\sigma = 80 \mu\Omega - mm$$

$$R_{Brass1} = \sigma \frac{l}{A} = 311.334 \mu\Omega$$

R2: - Terminal 2 resistance

$$A = 0.817mm^2$$

$$l = 3.18mm$$

$$\sigma = 80 \mu\Omega - mm$$

$$R_{Brass2} = \sigma \frac{l}{A} = 311.334 \mu\Omega$$

Total Terminal Resistance,

$$R_{BT} = 622.668 \mu\Omega$$

R3: - Resistance of DBC under T2

$$A = 1.26mm^2$$

$$l = 10.83mm$$

$$\sigma = 16.8 \mu\Omega - mm$$

$$R_3 = \sigma \frac{l}{A} = 144.44 \mu\Omega$$

R4: - Resistance of DBC under MOSFET 2

First half with cross-sectional area 1.26 mm^2

$$A = 1.26 \text{ mm}^2$$

$$l = 4.59 \text{ mm}$$

$$\sigma = 16.8 \text{ u}\Omega - \text{mm}$$

$$R_{4a} = \sigma \frac{l}{A} = 61.2 \text{ u}\Omega$$

First half with cross-sectional area 2.37 mm^2

$$A = 2.37 \text{ mm}^2$$

$$l = 7.84 \text{ mm}$$

$$\sigma = 16.8 \text{ u}\Omega - \text{mm}$$

$$R_{4a} = \sigma \frac{l}{A} = 55.57 \text{ u}\Omega$$

$$R_{\text{total}} = R_1 + R_2 + R_3 + R_4$$

$$R_{\text{total}} = 883.878 \text{ u}\Omega$$

Therefore, the total estimated interconnect resistance **883.878 uΩ** which is < 3% of the MOSFET's on-state resistance ($35 \text{ m}\Omega \times 0.03 = 1.05 \text{ m}\Omega$).

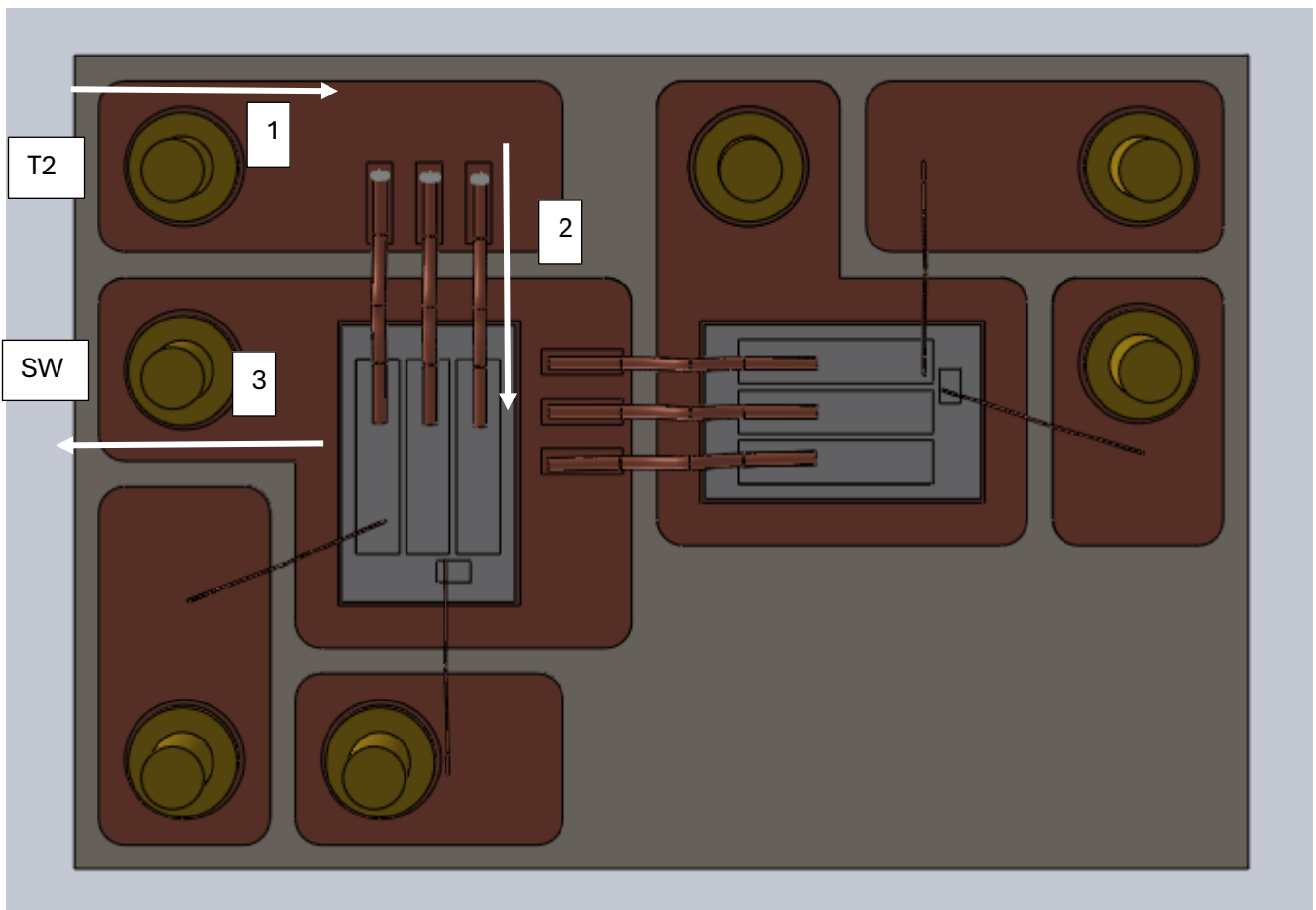


Figure 2.20 Freewheeling Path

Thermal Analysis

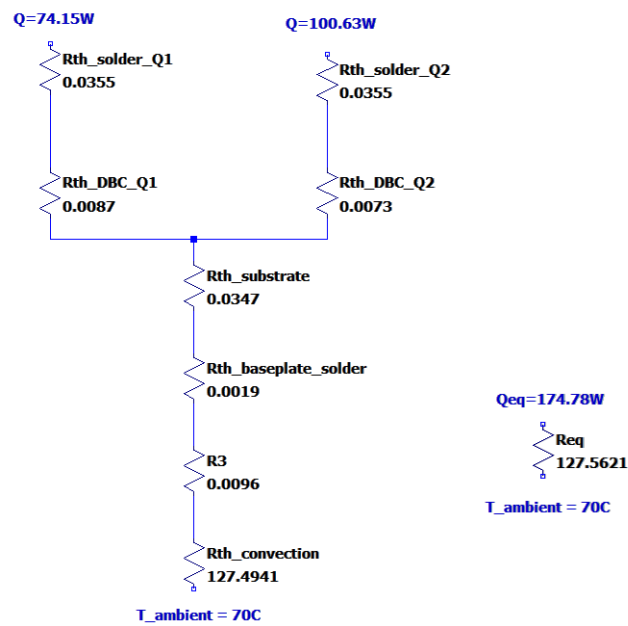


Figure 2.21 Equivalent Thermal Resistance

Figure 2.21 illustrates the full and equivalent thermal resistance models for the half bridge module design. The thermal resistances were calculated according to the material thermal conductivities and the design outlined in parts 2b and 2c. Due to a variation in DBC dimensions and losses, the two MOSFETs do not have symmetrical thermal profiles. Using the equivalent resistance model in Figure 2.21, junction temperature in the range of 22000°C can be calculated. This is obviously **far** above a rational design.

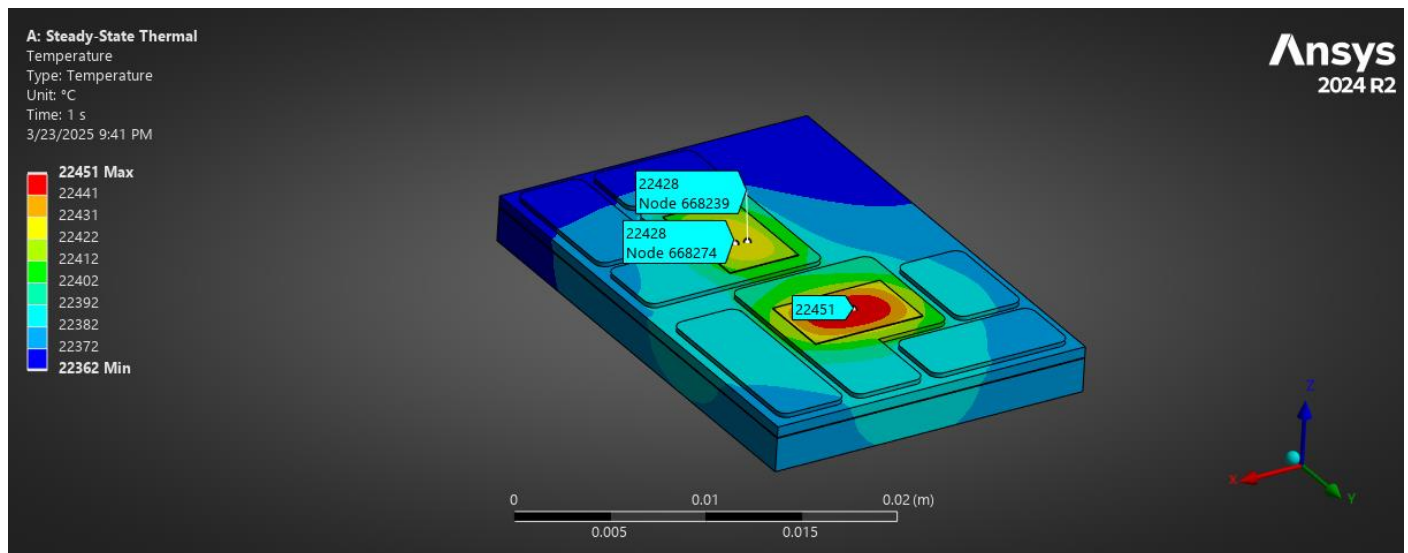


Figure 2.21 Junction Temperatures

To corroborate the junction temperatures calculated above, a steady-state thermal simulation was performed in Ansys. Figure 2.21 displays the results. Terminals and wire bonds were suppressed to simplify the simulation. The Ansys simulation resulted in similar junction temperatures as calculated above. Thermally, our design was a failure. The main culprit, in addition to the losses themselves, is the extremely low convective

Heat Rate, which is not far above that of standing air. As can be seen in Figure 2.20, the equivalent thermal resistance is essentially equal to that of the thermal resistance of convective heat transfer!

A few avenues were explored which could improve our design. While outside the scope of the assignment, we analysed the results of a larger baseplate area and a better heat rate h . Each of these variables affect the thermal resistance of convection. With $h=5000 \text{ W/m}^2\text{C}$, a value achievable with a well-designed liquid cooling system, the junction temperatures look more reasonable. However, they are still high.

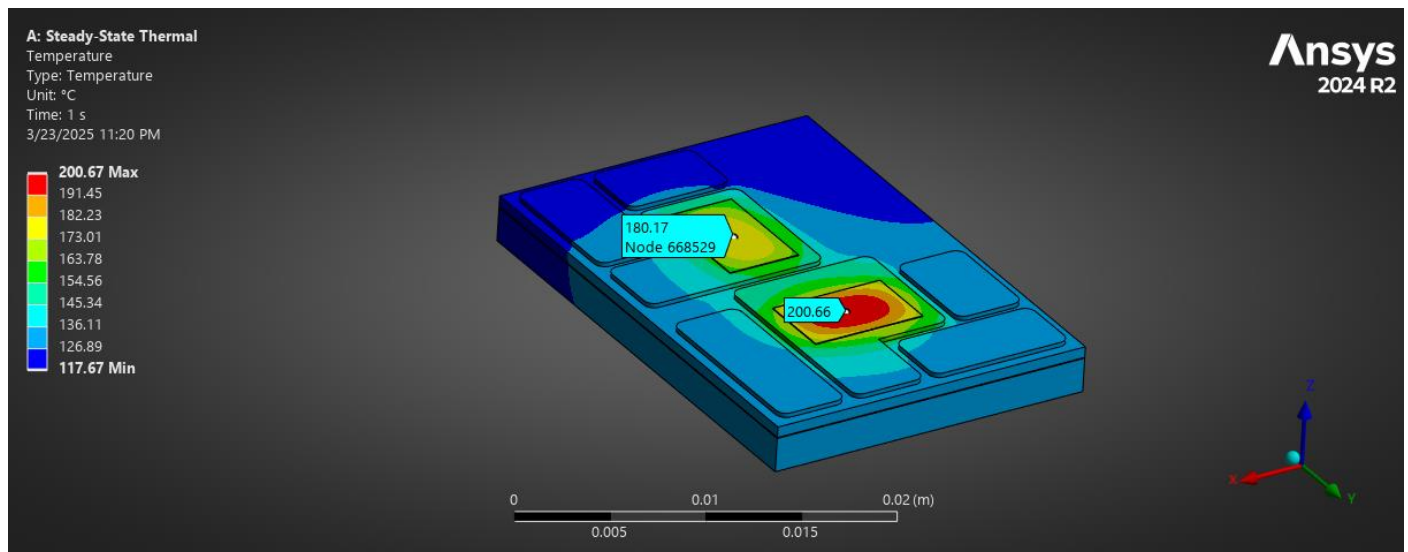


Figure 2.22 Junction Temperatures with $h=5500 \text{ W/m}^2\text{C}$

Another solution that does get a good junction temperature is increasing the baseplate size to somewhere in the range of $150,000 \text{ mm}^2$, decreasing the thermal resistance of convection to 0.4444 W/mK . This is unrealistic, however. Ultimately, the best, most realistic solution is to use a heat sink along with forced-air convective cooling. A heat sink can drastically increase the surface area without significantly increasing the baseplate size. A combination of baseplate size increase, heatsink addition, and increase airflow makes this design work. By increasing the baseplate by only 20% -- which is reasonable as we were previously targeting a relatively compact design -- and adding a heatsink that increases surface area by a factor of 10, then we need only a heat rate of $300 \text{ W/m}^2\text{K}$ to achieve a junction temperature of 175°C . This heat rate is very achievable with forced convection [5].

$$Q_{eq} = 174.78 \text{ W}$$

$$R_{eq} = 0.5992$$

$$T_{ambient} = 70^\circ\text{C}$$

Figure 2.23 Equivalent Thermal Resistance of improved design

While the above design is outside the scope of the assignment, it is possible without going to engineering extremes.

Problem 3: PCB-Based Planar Inductor

Problem Statement:

In this problem, we have to design a planar inductor and find the inductance and interwinding capacitance value at 1MHz frequency.

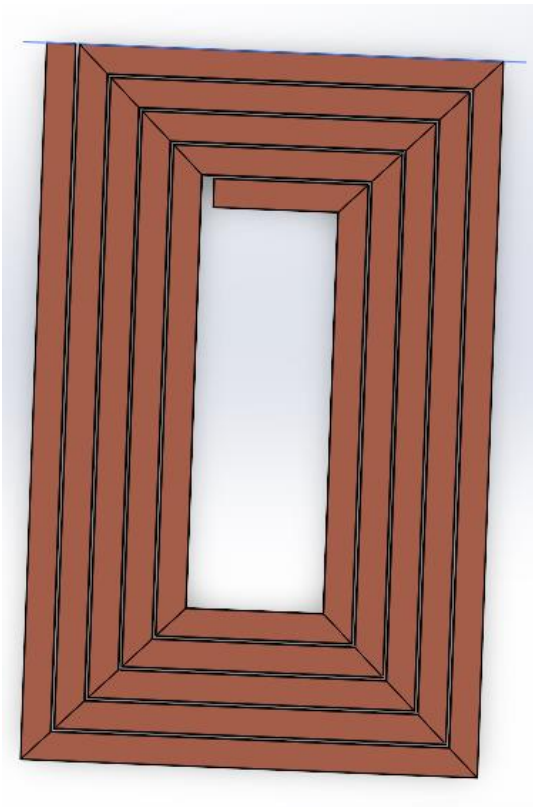
The design parameters are as follows:

- i) Number of turns in each winding: 5
- ii) $D_{in}=9.906\text{mm}$
- iii) Copper trace winding thickness on each side: $35\mu\text{m}$
- iv) ARC layer thickness: $40\mu\text{m}$
- v) EI core material: 3F4
- vi) Frequency: 1MHz
- vii) Gap between traces: 0.251mm
- viii) Copper trace width: 2.1mm
- ix) Air gap length in the magnetic path: 0.32mm

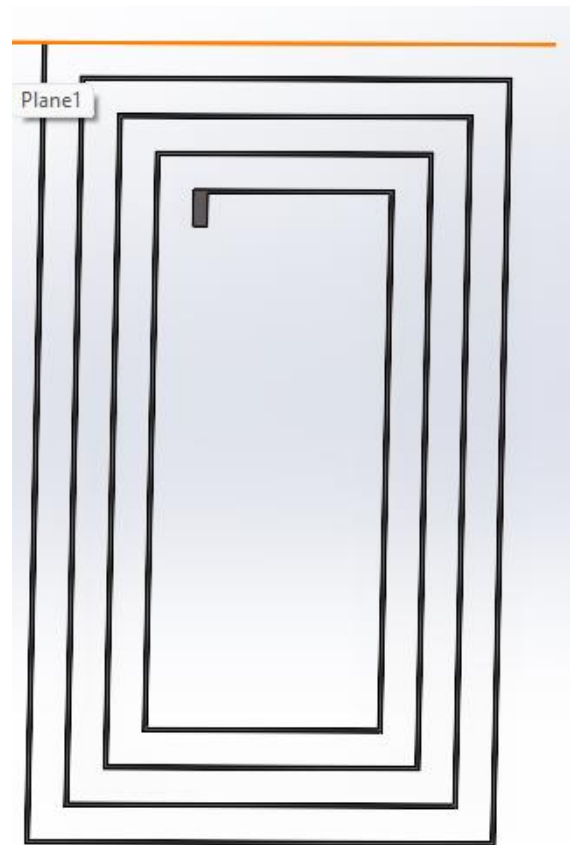
Part 3a Design

From the dimensional constraint of EI core [1] and some design parameters from [2], dimensions of upper copper winding, lower copper windings and the ARC layer design were determined. Then each component parts were designed and assembled in Solid Works. The assembled CAD model was then imported into ANSYS Q3D extractor. Materials were assigned. Some of the materials like 3F4 (EI-core) and Risho Molding Compound (filling in the gaps between the traces) and the Aluminium Ceramic (material properties were modified as provided to us in the software) were custom added using the material parameters provided and from the 3F4 datasheet. Analysis was performed.

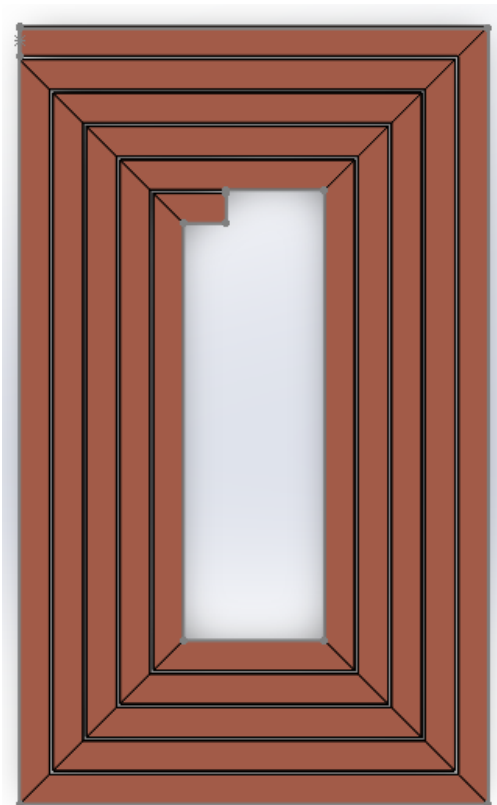
1)Copper Winding1:



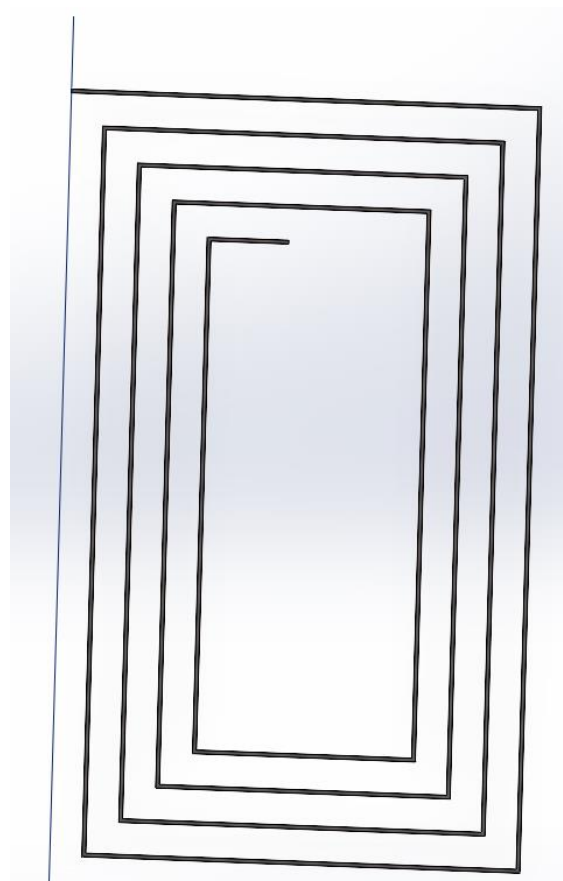
2) Copper winding1 Risho Molding



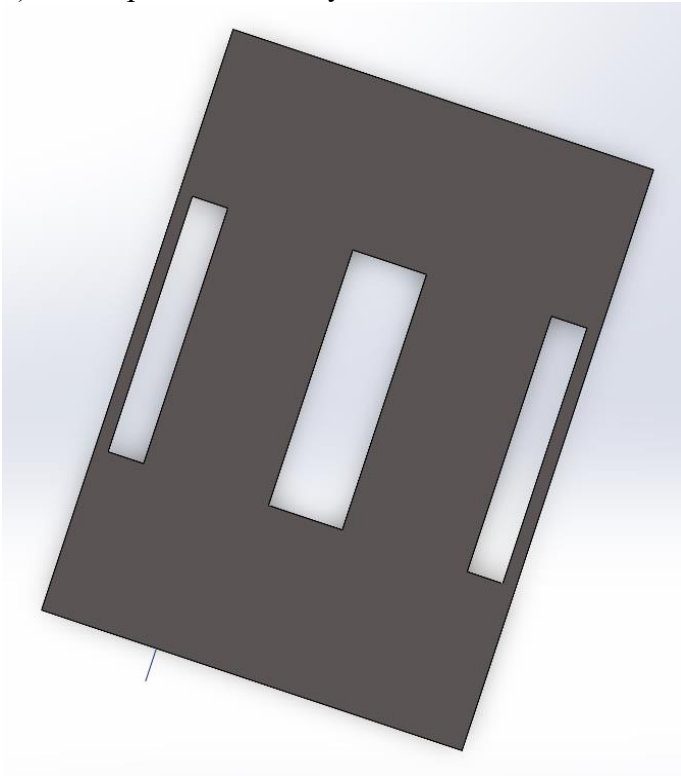
3)Copper winding2:



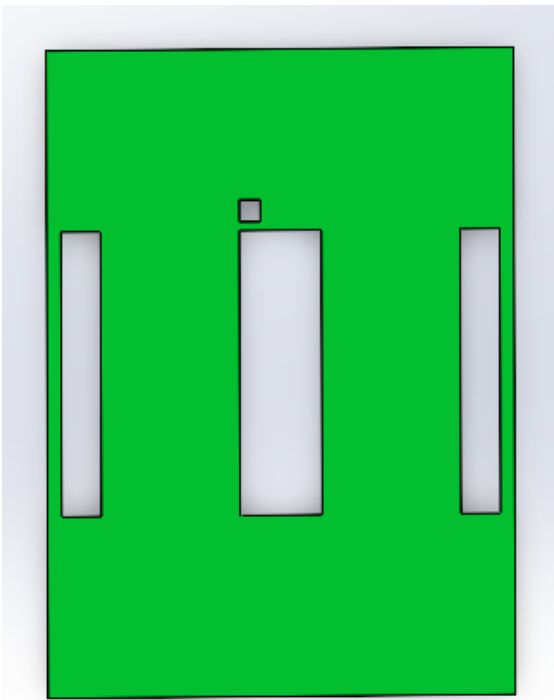
4) Copper winding2 Risho Molding:



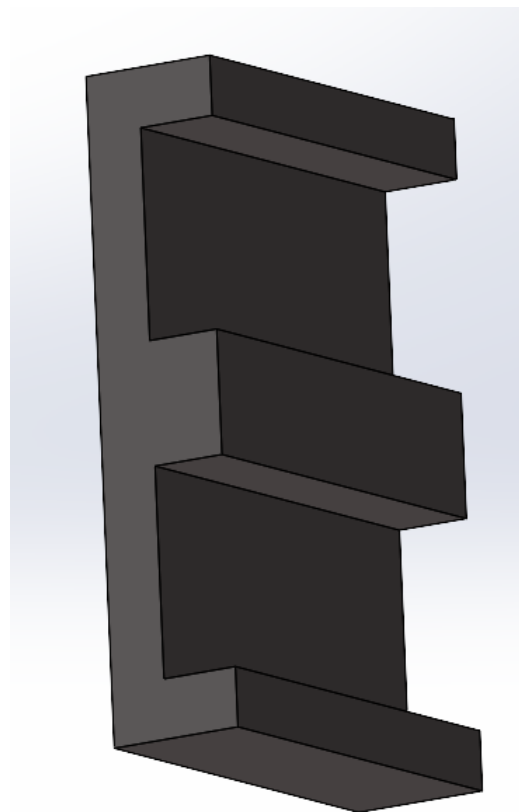
5)ARC top and bottom layer:



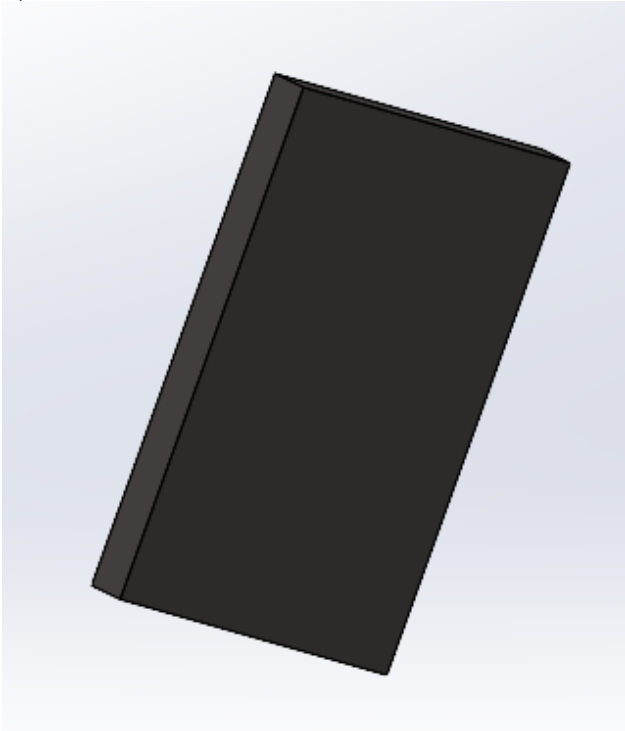
6)ARC middle layer with hole:



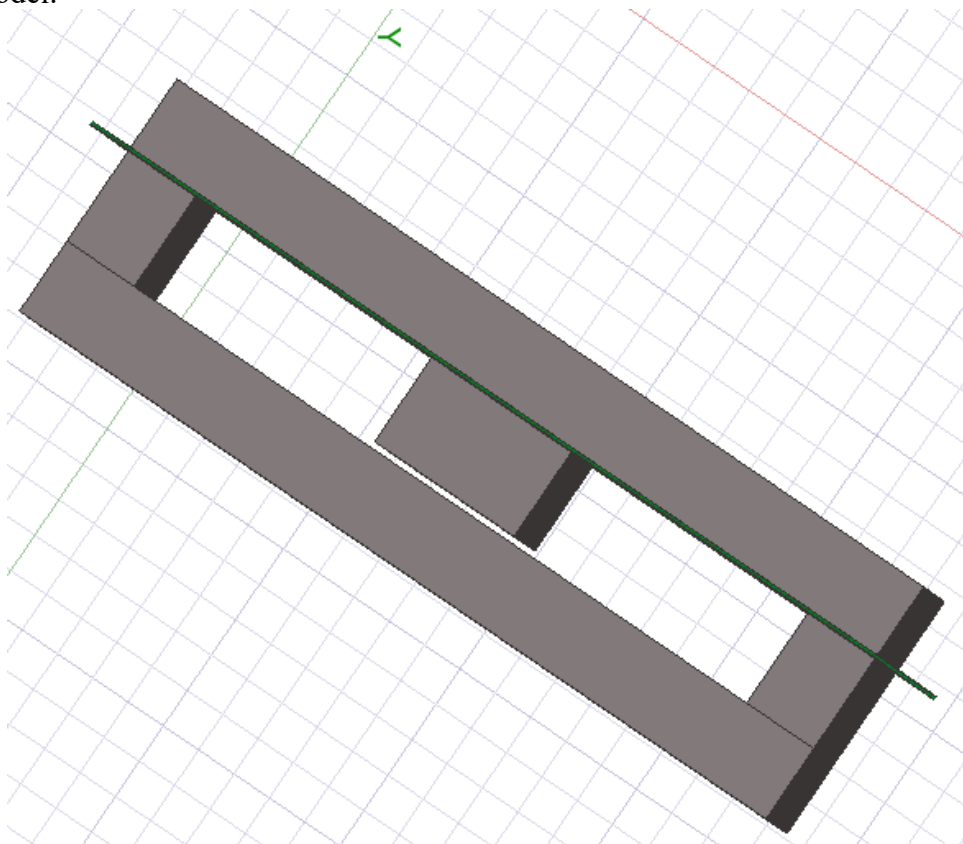
7) E core:

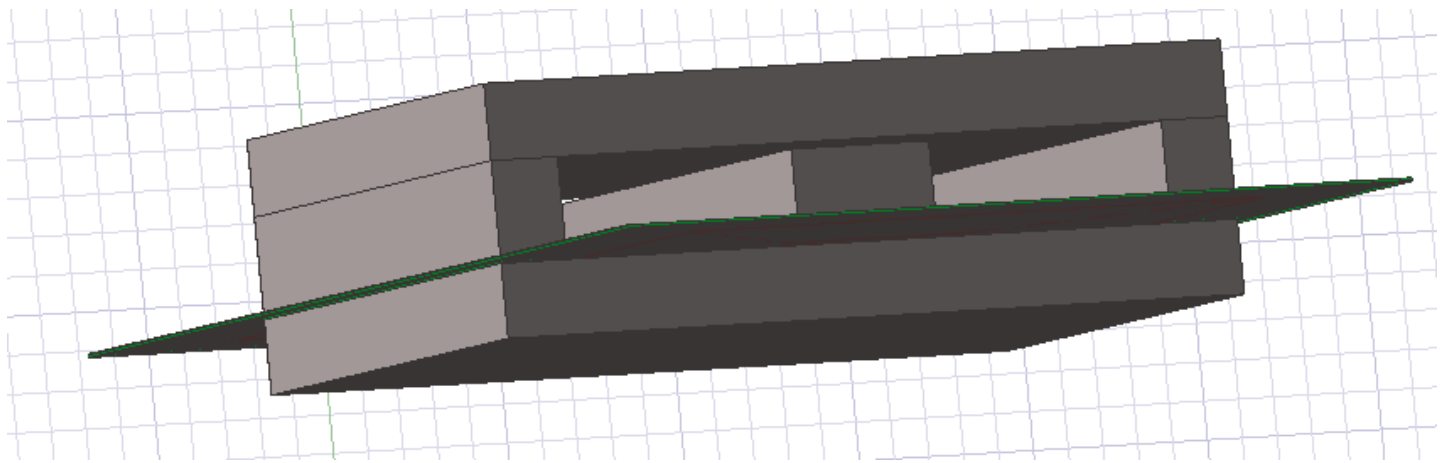


8) I core:



9) Assembled Model:





Part 3b Results

Inductance value:

Solutions: ece533_problem3 (1) - Q3DDesign1

Simulation: Setup1 LastAdaptive AC RL

Design Variation:

Profile Convergence Matrix Mesh Statistics

☐ Resistance Units: ohm Matrix 1MHz

☒ Inductance Units: uH Original ☐ All Freqs

☐ Self Terms

View Format Passivity Export

	c1:Source1
Freq: 1MHz	
c1:Source1	7.32252

Close

Capacitance value:

Solutions: ece533_problem3 (1) - Q3DDesign1

Simulation: Setup1 LastAdaptive CG

Design Variation:

Profile Convergence Matrix Mesh Statistics

☐ Conductance Units: mSie Maxwell Matrix 1MHz

☒ Capacitance Units: pF Original ☐ All Freqs

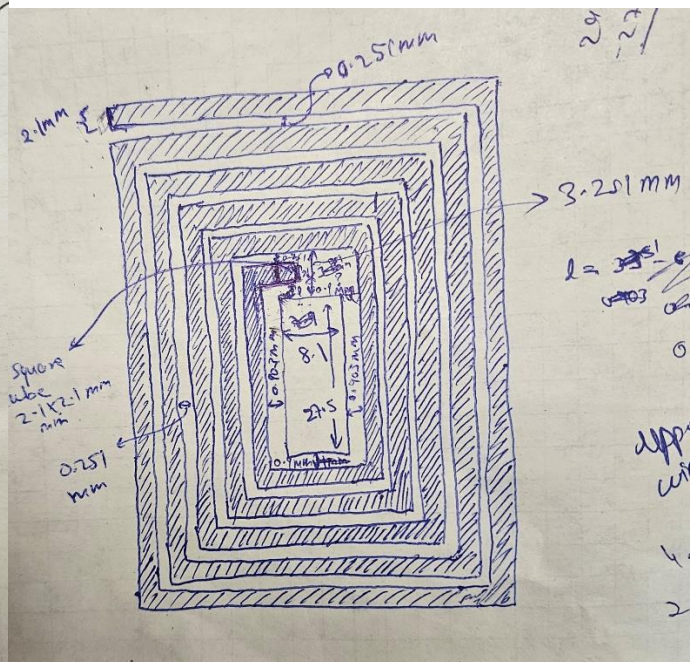
☐ Self Terms

View Format Passivity Export

	c1
Freq: 1MHz	
c1	1.36758

Close

1) Multiple attempts of redesigning and remodelling copper windings, respective Risho molding and readjusting dimensions of Arc layer after realizing design constraint getting violated. Some of the initial unsuccessful designs, the copper windings thought to be just mirror image of one another.



2) Multiple attempts of assembling all the parts due to mismatch in mating

Reference

- [1] PCB-Based Inductor Design for 1-Kw, 1-MHz Buck Converter
- [2] FERROXCUBE E43/10/28 + PLT43/28/4.1
- [3] Wolf-speed, “Silicon Carbide Power MOSFET Bare Die”, CPM2-1200-0025B
- [4] Lect-12 Intro and Design Guides for Direct Bonded Copper Substrates & Wire Bonding by Dr. Hopkins
- [5] Understanding Convective Heat Transfer: Coefficients, Formulas & Examples, The Engineering Toolbox